

CME 296: Diffusion & Large Vision Models



Afshine Amidi & Shervine Amidi





Diffusion & Large Vision Models

Recap

State-of-the-art models

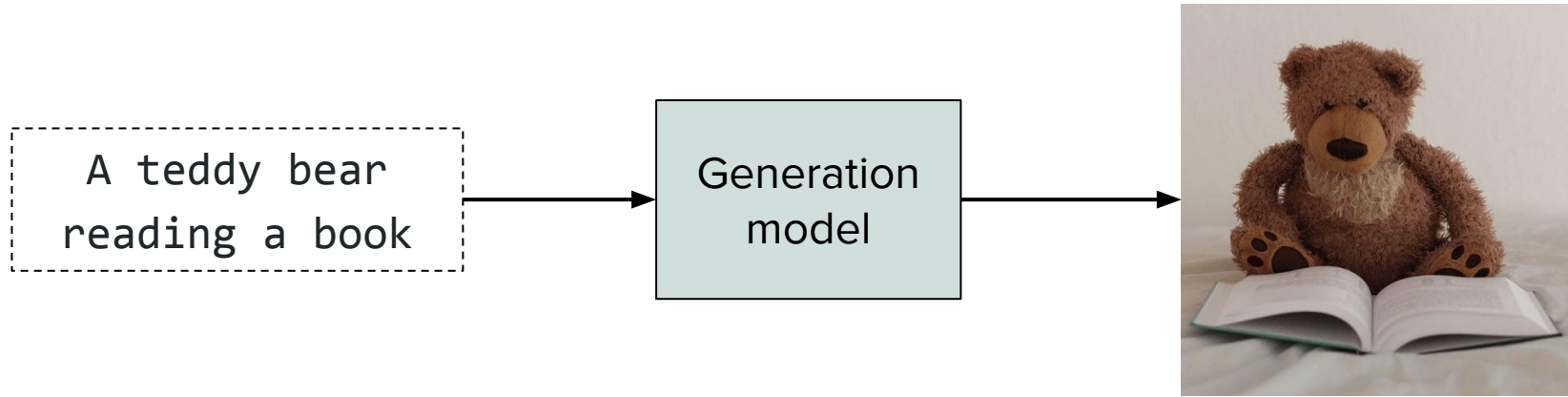
Extension to videos

Alternatives to generation

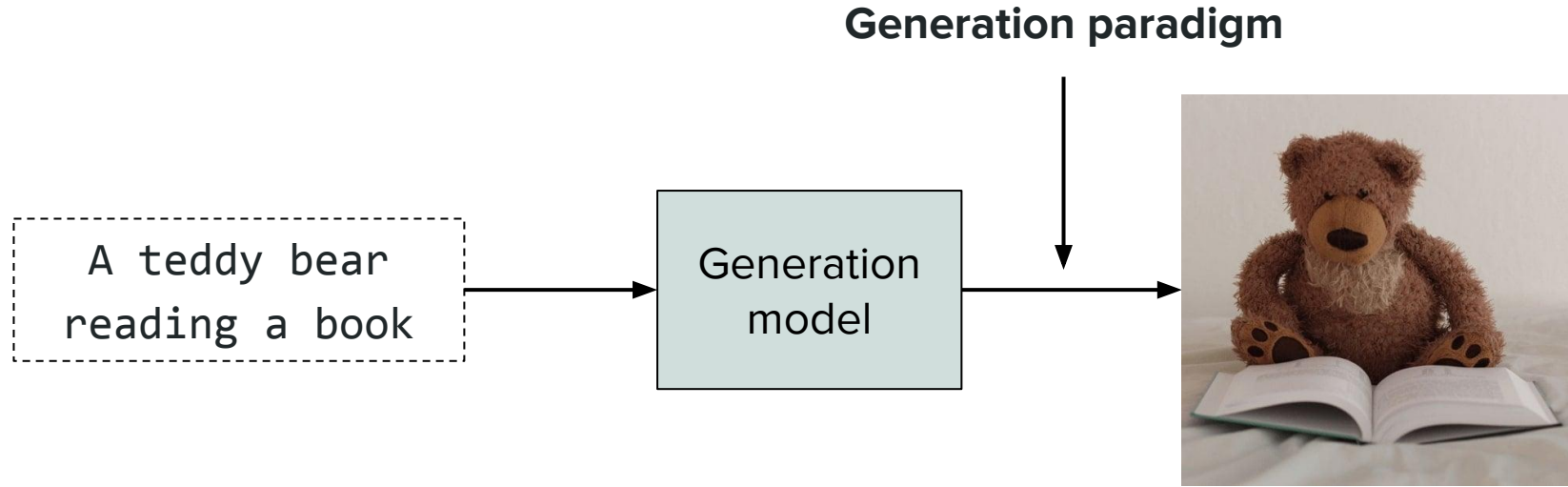
Diffusion for LLMs

Closing thoughts

Rewinding the quarter...



Rewinding the quarter...



Rewinding the quarter...

Lecture 1

Diffusion

Rewinding the quarter...

Lecture 1

Forward process

$$q(x_t|x_{t-1})$$

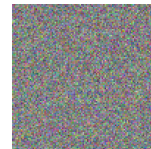
$$x_{\text{image}} = x_0$$



...



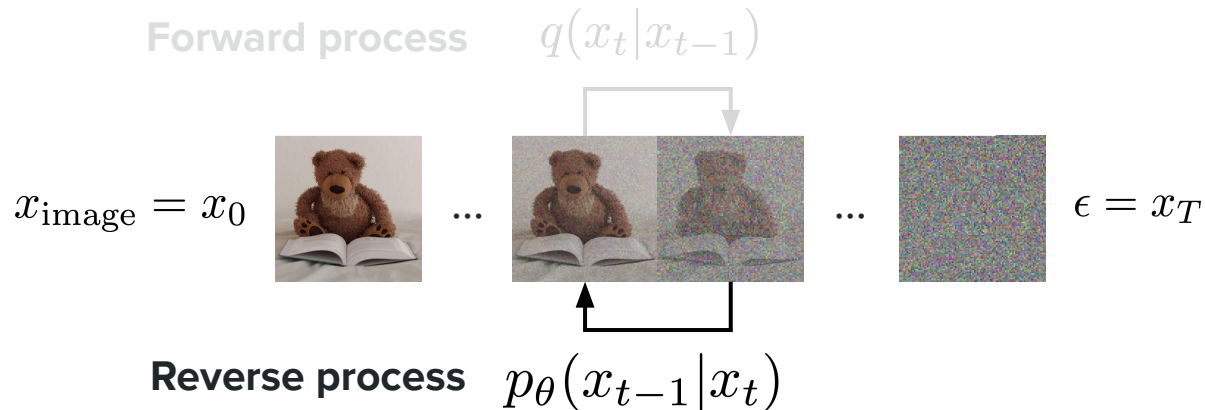
...



$$\epsilon \approx x_T$$

Rewinding the quarter...

Lecture 1



Rewinding the quarter...

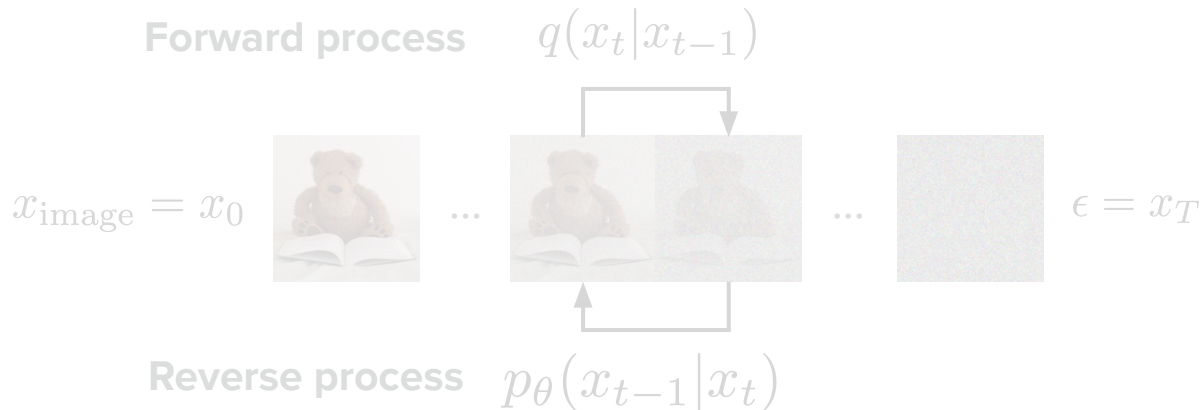
Lecture 1

1 Derive **lower bound**

2 **Expand** terms

3 Show **tractable** bound

4 Deduce **final loss**



Rewinding the quarter...

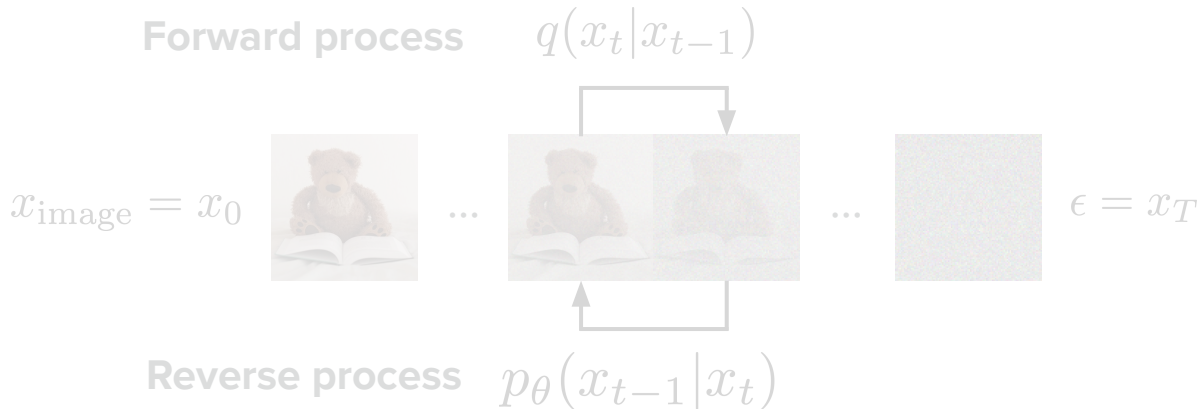
Lecture 1

1 Derive **lower bound**

2 **Expand** terms

3 Show **tractable** bound

4 Deduce **final loss**



Noise predictor

Actual noise

$$\mathcal{L} = \mathbb{E}_{t, x_0, \epsilon} \left[\left\| \epsilon_\theta \left(\frac{\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t \right) - \epsilon \right\|^2 \right]$$

Noised image

Noise level

Rewinding the quarter...

Lecture 1

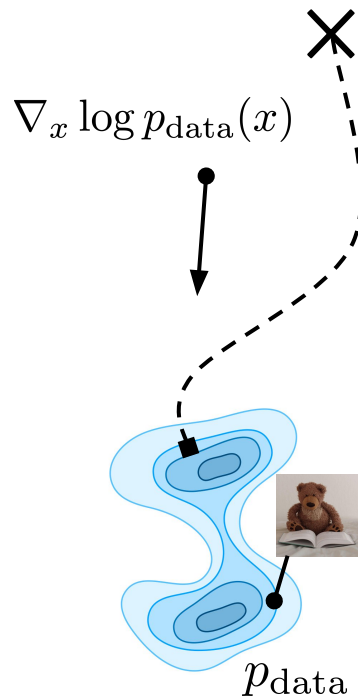
Lecture 2

Score matching

Rewinding the quarter...

Lecture 1

Lecture 2

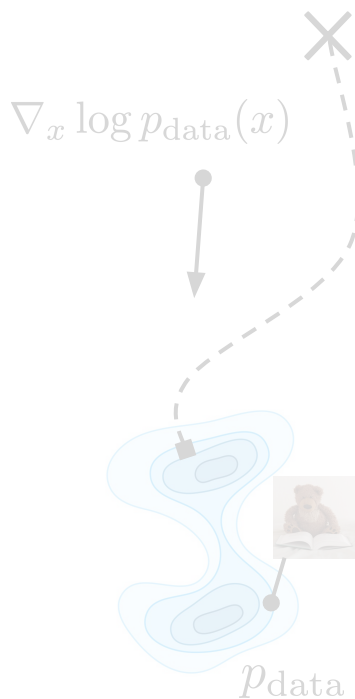


Rewinding the quarter...

Lecture 1

Lecture 2

$$\mathcal{L}_{\text{DSM}} = \mathbb{E}_{t, x_0, x_t} \left[\lambda_t \|s_\theta(x_t, t) - \nabla_{x_t} \log p(x_t | x_0)\|^2 \right]$$

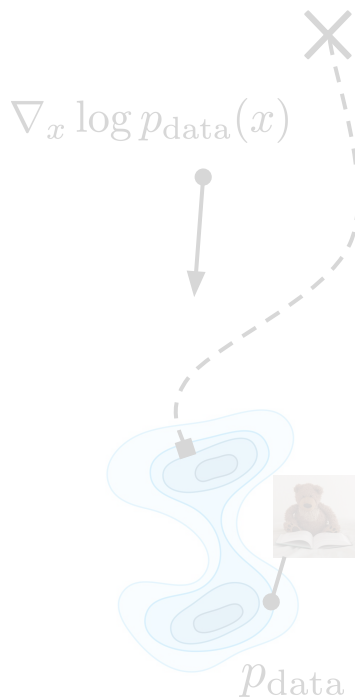


Rewinding the quarter...

Lecture 1

Lecture 2

$$\mathcal{L}_{\text{DSM}} = \mathbb{E}_{t, x_0, x_t} \left[\lambda_t \|s_\theta(x_t, t) - \nabla_{x_t} \log p(x_t | x_0)\|^2 \right]$$



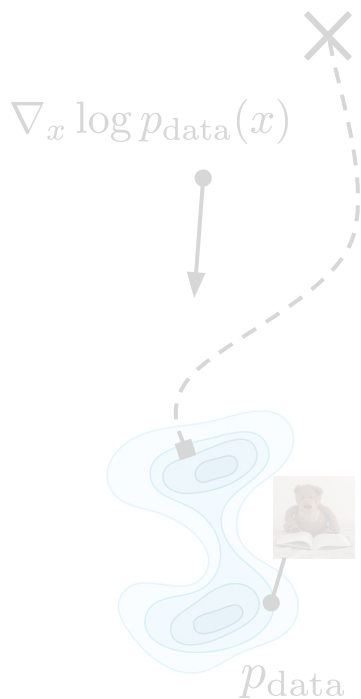
Score from noisy
to clean $\longrightarrow \nabla_{x_t} \log(q(x_t | x_0)) = -\frac{\epsilon}{\sqrt{1 - \bar{\alpha}_t}} \longleftarrow$ **Noise** to remove
from noisy sample

Rewinding the quarter...

Lecture 1

Lecture 2

$$\mathcal{L}_{\text{DSM}} = \mathbb{E}_{t, x_0, x_t} \left[\lambda_t \| s_\theta(x_t, t) - \nabla_{x_t} \log p(x_t | x_0) \|^2 \right]$$



Score from noisy to clean $\rightarrow \nabla_{x_t} \log(q(x_t | x_0)) = -\frac{\epsilon}{\sqrt{1 - \bar{\alpha}_t}} \leftarrow$ Noise to remove from noisy sample

Forward $dx = \underbrace{f(x, t)dt}_{\text{Drift}} + \underbrace{g(t)dW}_{\text{Diffusion}}$

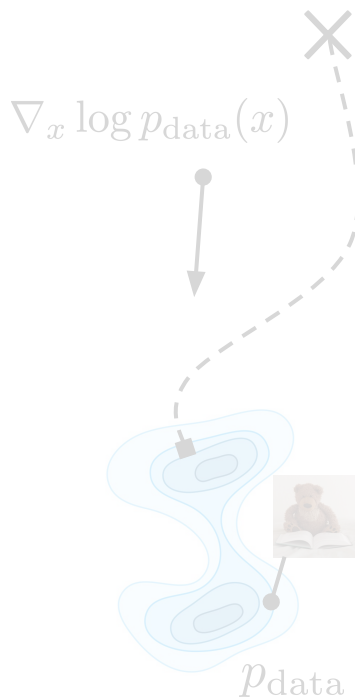
Diagram illustrating the forward process. A point x is transformed to $x + dx$ via the forward process. The transformation is decomposed into a drift component $f(x, t)dt$ (dashed blue line) and a diffusion component $g(t)dW$ (dashed red line).

Rewinding the quarter...

Lecture 1

Lecture 2

$$\mathcal{L}_{\text{DSM}} = \mathbb{E}_{t, x_0, x_t} \left[\lambda_t \|s_\theta(x_t, t) - \nabla_{x_t} \log p(x_t | x_0)\|^2 \right]$$



Score from noisy to clean $\rightarrow \nabla_{x_t} \log(q(x_t | x_0)) = -\frac{\epsilon}{\sqrt{1 - \bar{\alpha}_t}} \leftarrow$ Noise to remove from noisy sample

Forward $dx = \underbrace{f(x, t)dt}_{\text{Drift}} + \underbrace{g(t)dW}_{\text{Diffusion}}$

Diagram illustrating the forward process. A solid arrow represents the drift term $f(x, t)dt$, and a dashed arrow represents the diffusion term $g(t)dW$. The process starts at x and ends at $x + dx$.

Reverse $dx = \left[f(x, t) - g(t)^2 \nabla_x \log(p_t(x)) \right] dt + g(t)d\bar{W}$

Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

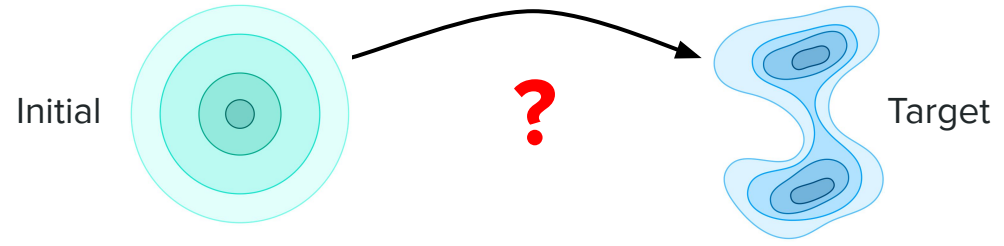
Flow matching

Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3



Rewinding the quarter...

Lecture 1

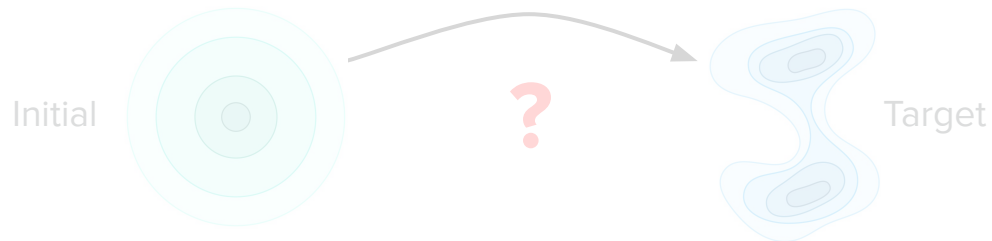
Lecture 2

Lecture 3

Single
sample

$$\frac{dx}{dt} = u_t(x)$$

$$\begin{array}{ccc} & u_t(x)dt & \\ & \xrightarrow{\quad} & \\ x & & x + dx \end{array}$$



Rewinding the quarter...

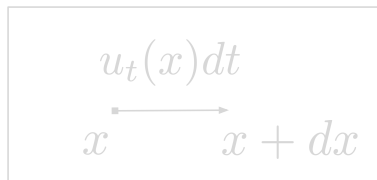
Lecture 1

Lecture 2

Lecture 3

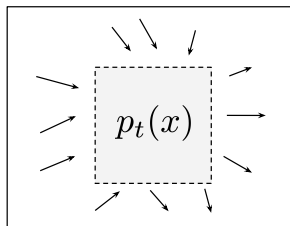
Single
sample

$$\frac{dx}{dt} = u_t(x)$$



$$\frac{\partial p_t}{\partial t}(x) = -\nabla \cdot (p_t u_t)(x)$$

Distribution



Initial



?

Target



Rewinding the quarter...

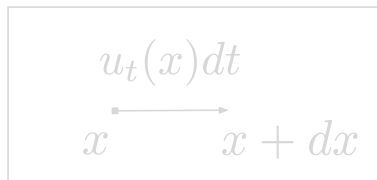
Lecture 1

Lecture 2

Lecture 3

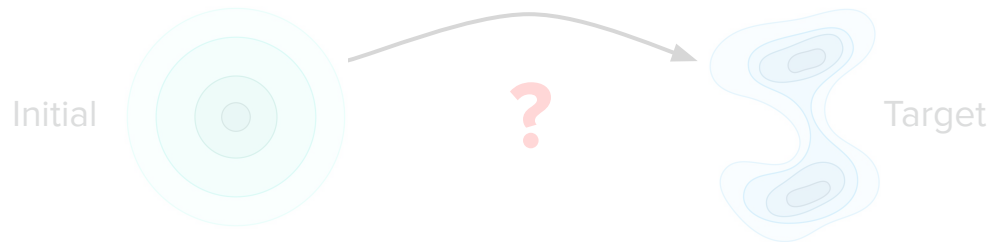
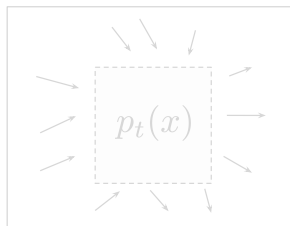
Single
sample

$$\frac{dx}{dt} = u_t(x)$$



$$\frac{\partial p_t}{\partial t}(x) = -\nabla \cdot (p_t u_t)(x)$$

Distribution



- 1 **Training.** Estimate $u_t(x)$ via $u_t^\theta(x)$
- 2 **Inference.** Map $x_0 \sim p_0$ to $x_1 \sim p_1$ via $u_t^\theta(x)$

$$\hat{x}_1 = x_0 + \int_0^1 u_t^\theta(x) dt$$

Rewinding the quarter...

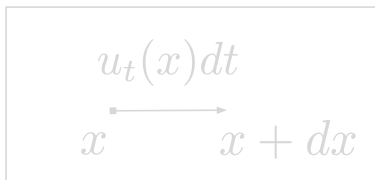
Lecture 1

Lecture 2

Lecture 3

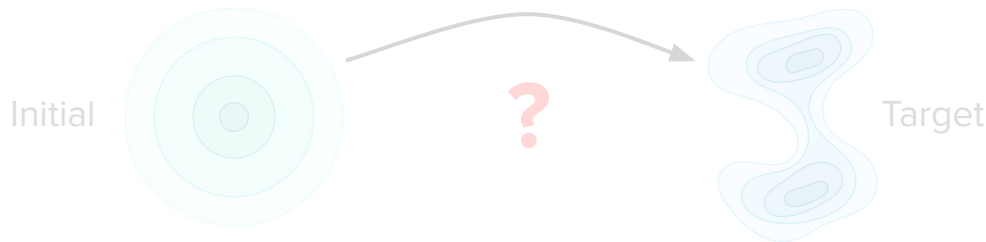
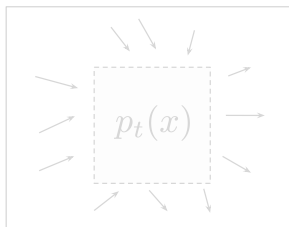
Single
sample

$$\frac{dx}{dt} = u_t(x)$$



$$\frac{\partial p_t}{\partial t}(x) = -\nabla \cdot (p_t u_t)(x)$$

Distribution



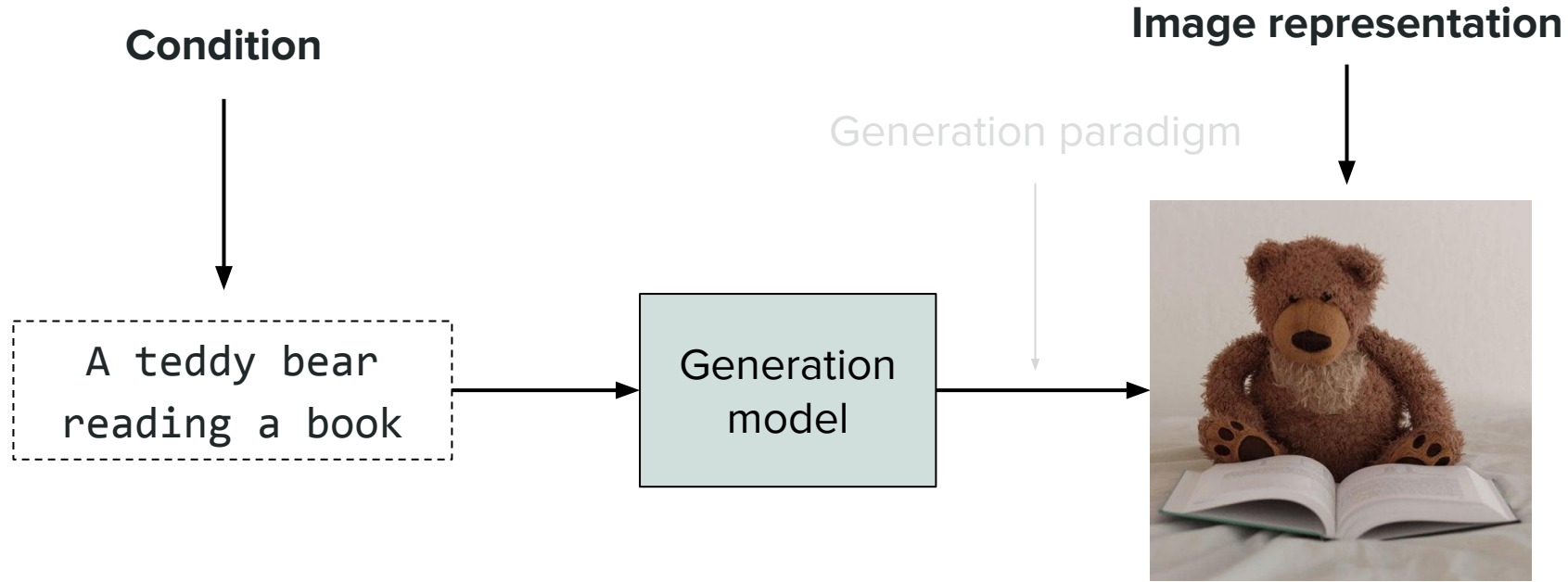
1 **Training.** Estimate $u_t(x)$ via $u_t^\theta(x)$

2 **Inference.** Map $x_0 \sim p_0$ to $x_1 \sim p_1$ via $u_t^\theta(x)$

$$\hat{x}_1 = x_0 + \int_0^1 u_t^\theta(x) dt$$

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, x_1, x} \left[\left\| u_t^\theta(x) - \overbrace{u_t(x \mid x_1)}^{x_1 - x_0} \right\|^2 \right]$$

Rewinding the quarter...



Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4

Latent space and guidance

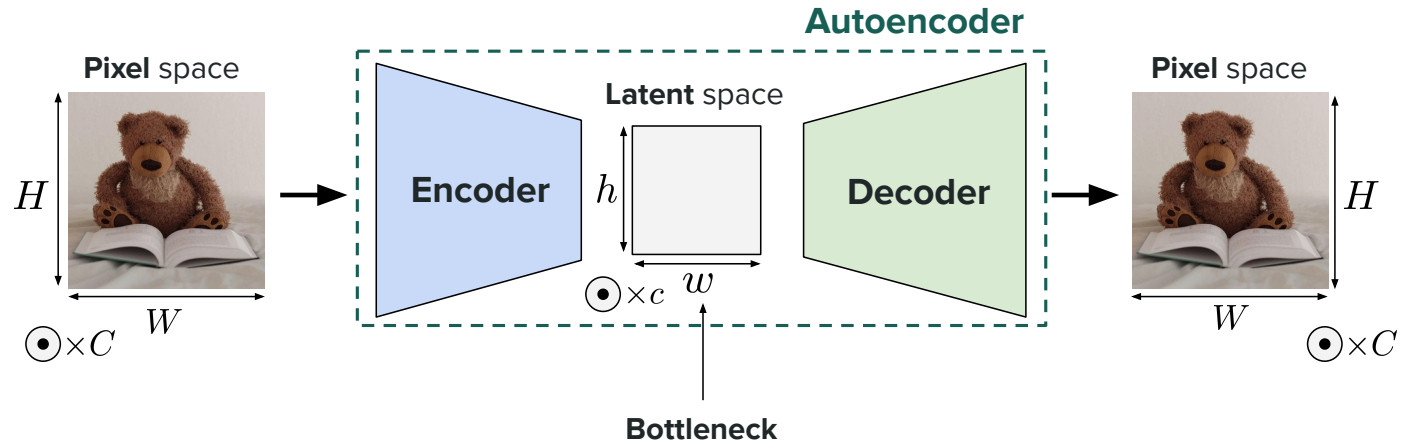
Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4



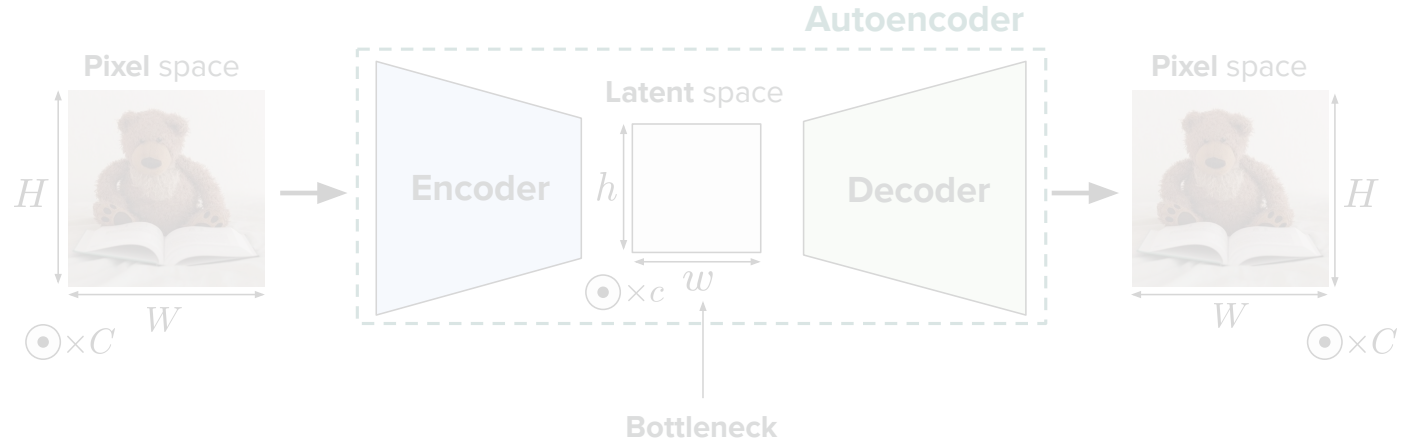
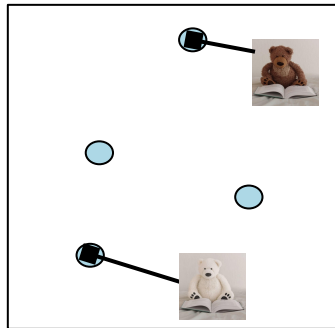
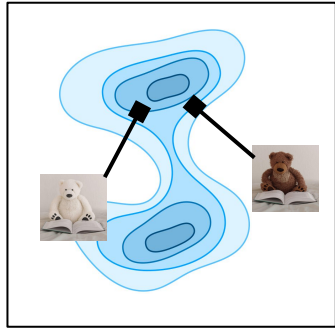
Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4



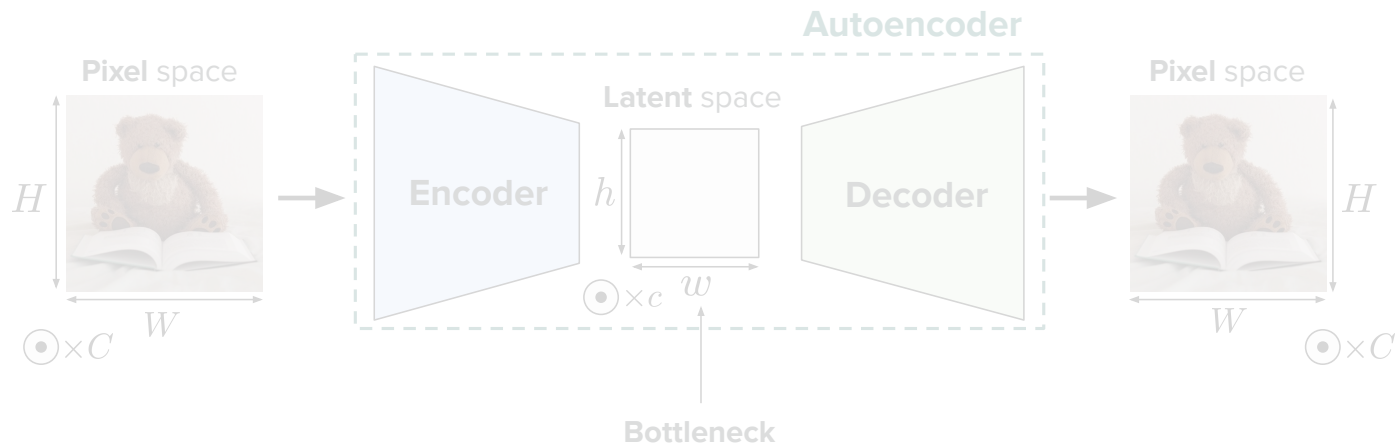
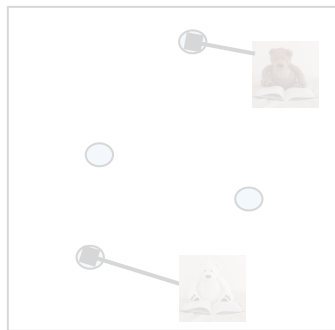
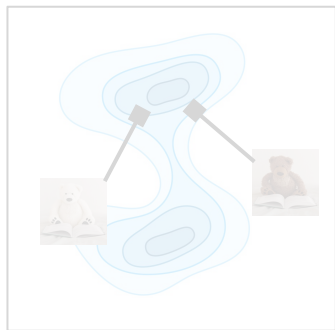
Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4

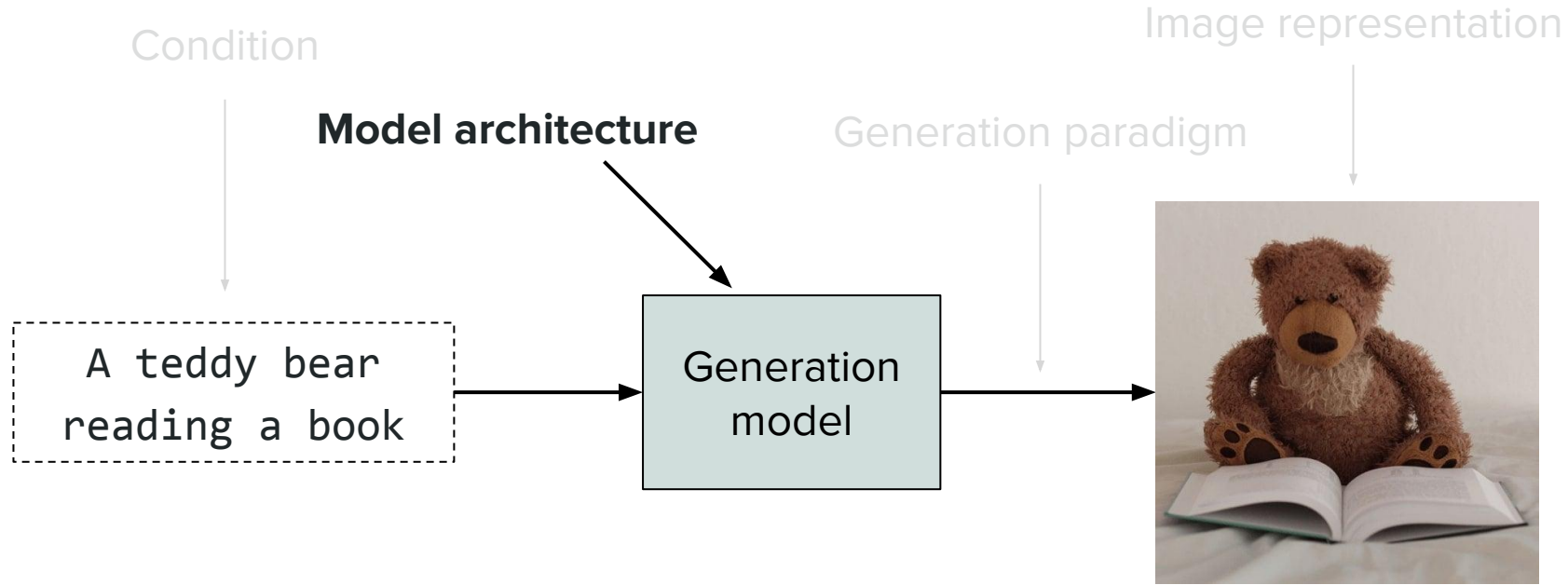


$$\mathcal{L}_{\text{VAE}} = \underbrace{-\mathbb{E}_z[\log p_\theta(x|z)]}_{\mathcal{L}_{\text{rec}}} + \underbrace{\text{KL}(q_\varphi(z|x) || p(z))}_{\mathcal{L}_{\text{KL}}}$$

Reconstruction

Regularization of latent space

Rewinding the quarter...



Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4

Lecture 5

Image generation architectures

Rewinding the quarter...

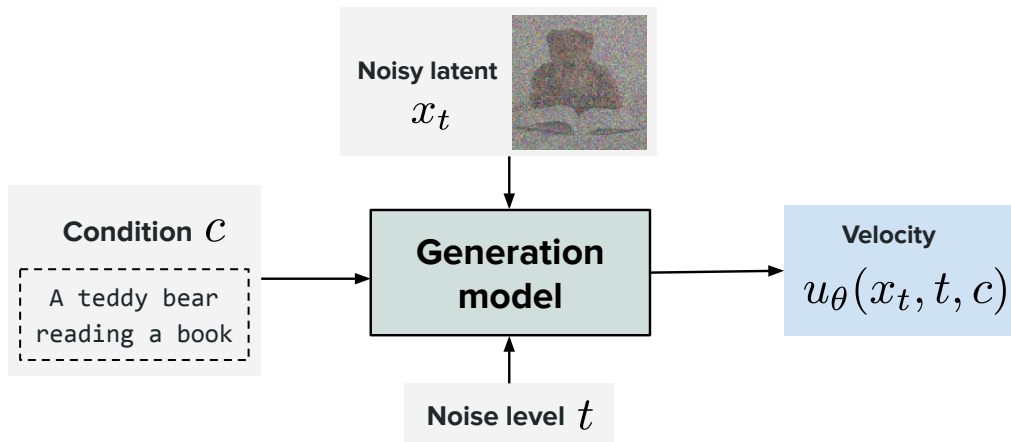
Lecture 1

Lecture 2

Lecture 3

Lecture 4

Lecture 5



Rewinding the quarter...

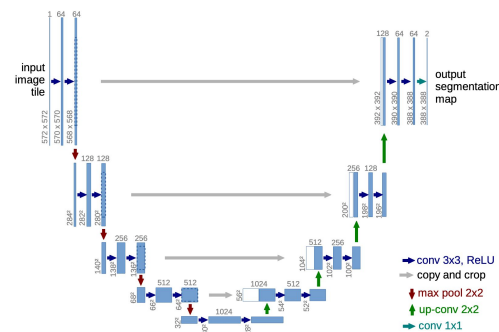
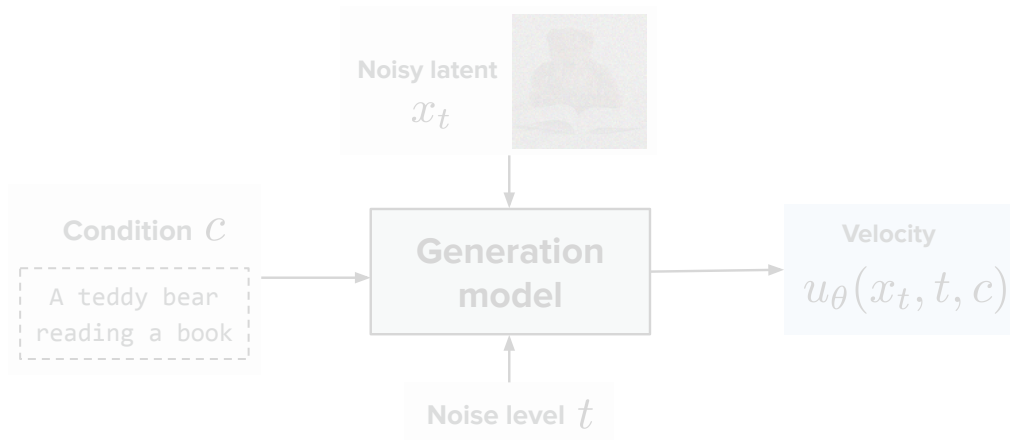
Lecture 1

Lecture 2

Lecture 3

Lecture 4

Lecture 5



U-Net
& variants

Lecture 5

Rewinding the quarter...

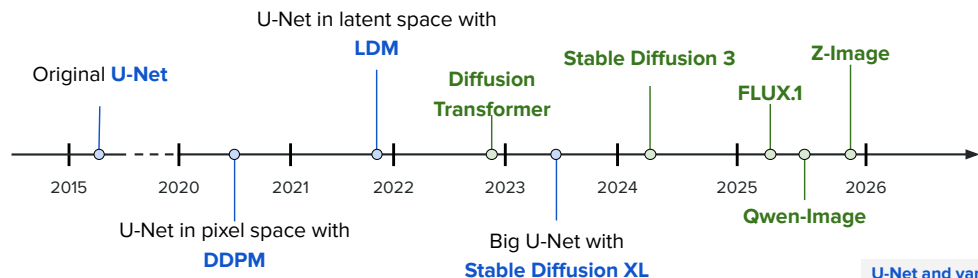
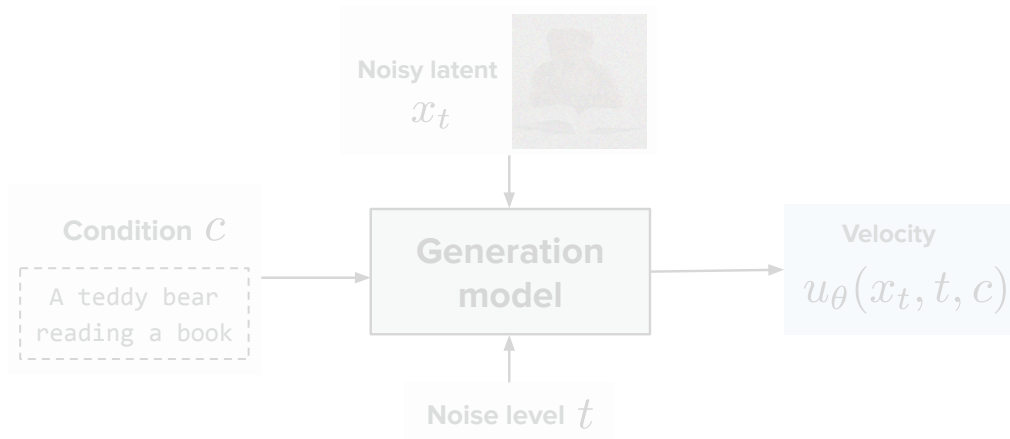
Lecture 1

Lecture 2

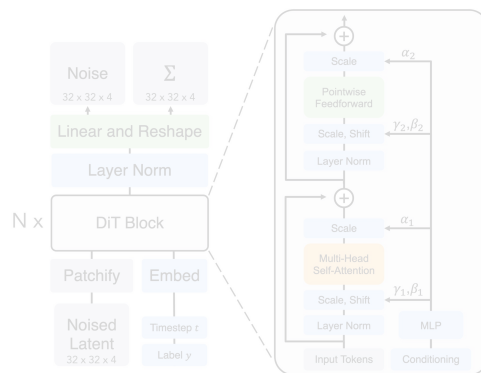
Lecture 3

Lecture 4

Lecture 5

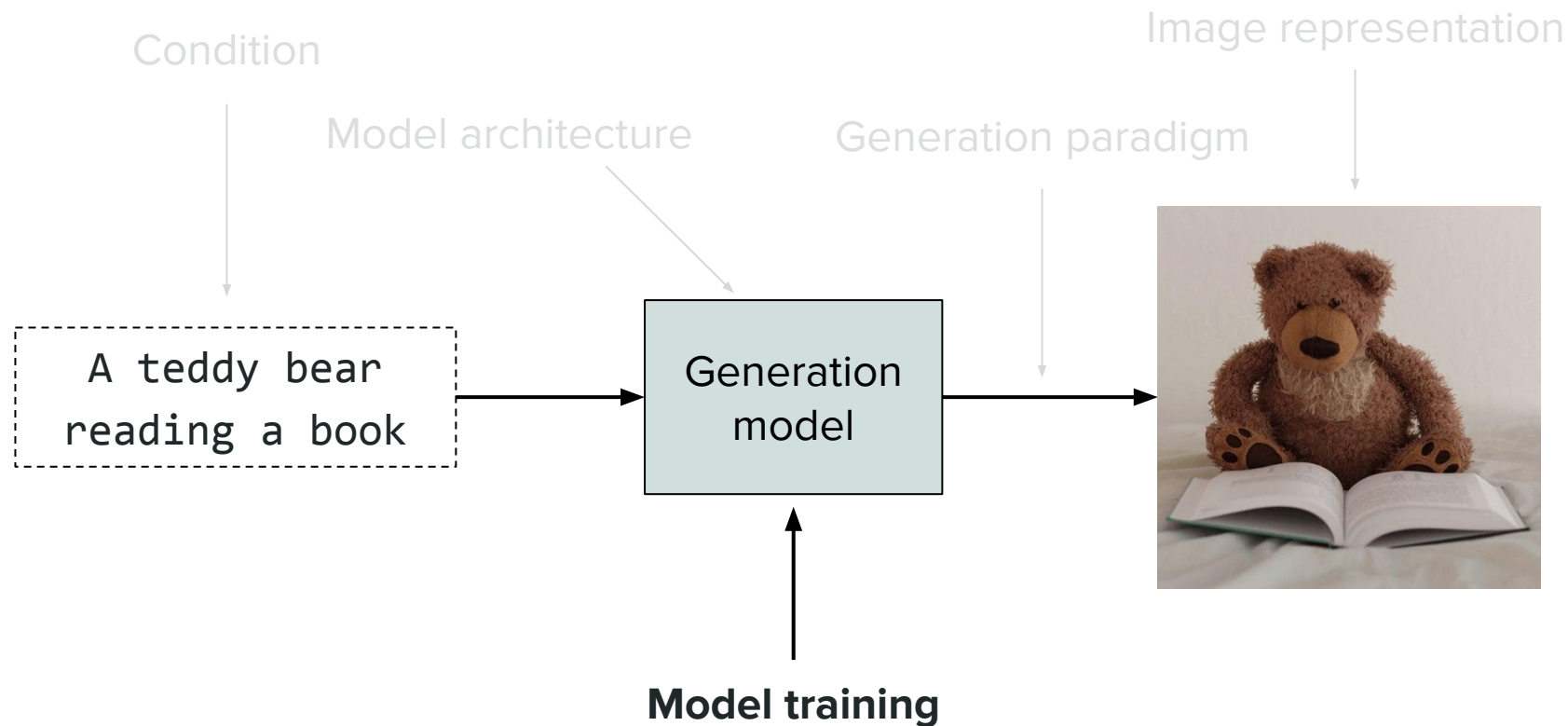


U-Net
& variants



Diffusion
Transformer
& variants

Rewinding the quarter...



Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4

Lecture 5

Lecture 6

Model training

Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4

Lecture 5

Lecture 6

Pre-training



Learn **how** to
generate images

Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4

Lecture 5

Lecture 6

Pre-training



Learn **how** to
generate images

Post-training



Learn how to
generate **good**
images

Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4

Lecture 5

Lecture 6

Pre-training



Learn **how** to
generate images

Post-training



Learn how to
generate **good**
images

Tuning



Learn how to
generate images
for a special case

Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4

Lecture 5

Lecture 6

Pre-training



Learn **how** to
generate images

Post-training



Learn how to
generate **good**
images

Tuning



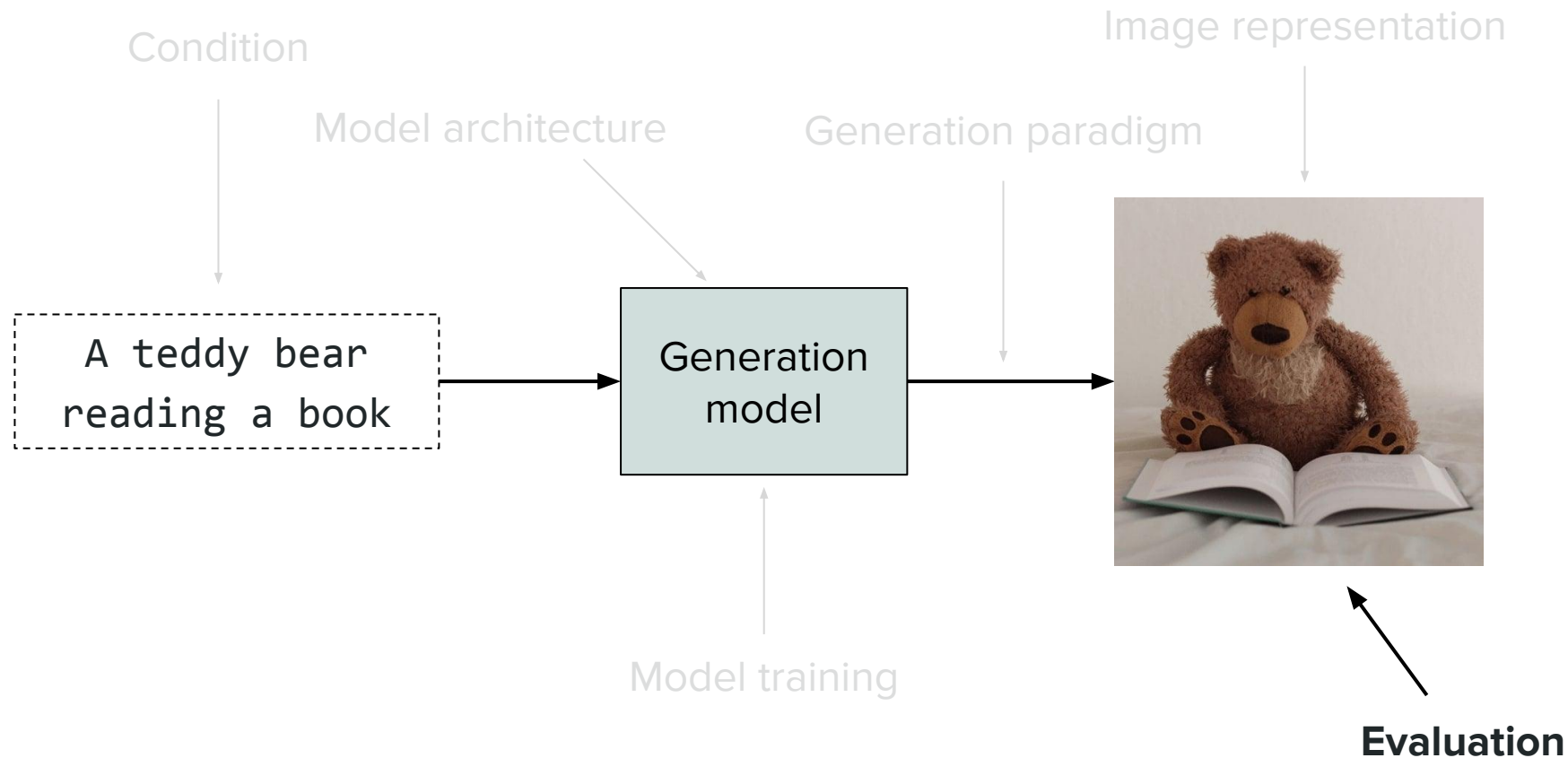
Learn how to
generate images
for a special case

Distillation



Learn how to
generate images,
fast

Rewinding the quarter...



Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4

Lecture 5

Lecture 6

Lecture 7

Evaluation

Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4

Lecture 5

Lecture 6

Lecture 7

■ Artificial Analysis Text to Image Leaderboard Artificial Analysis

Rank	Range	Creator	Model	Elo
1	1	OpenAI	GPT Image 2 (high)	1,338
2	2-3	OpenAI	GPT Image 1.5 (high)	1,266
3	2-3	Google	Nano Banana 2 (Gemini 3.1 Flash Image Preview)	1,264

$$R_A \leftarrow R_A + K \times \Delta$$

Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4

Lecture 5

Lecture 6

Lecture 7

Artificial Analysis Text to Image Leaderboard Artificial Analysis

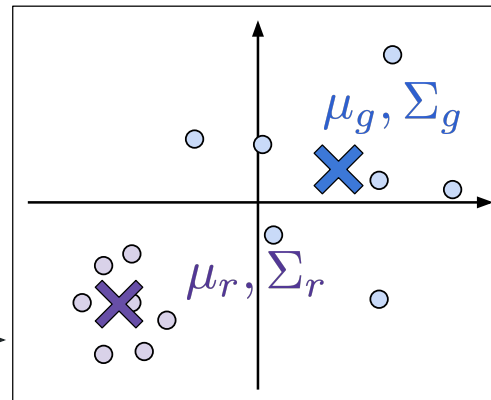
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$$R_A \leftarrow R_A + K \times \Delta$$

Real image
distribution



Pre-trained
encoder



Generated image
distribution



Pre-trained
encoder

Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4

Lecture 5

Lecture 6

Lecture 7

Artificial Analysis Text to Image Leaderboard

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$$R_A \leftarrow R_A + K \times \Delta$$

How cute is this teddy bear?



MLLM

Very cute!

Real image distribution

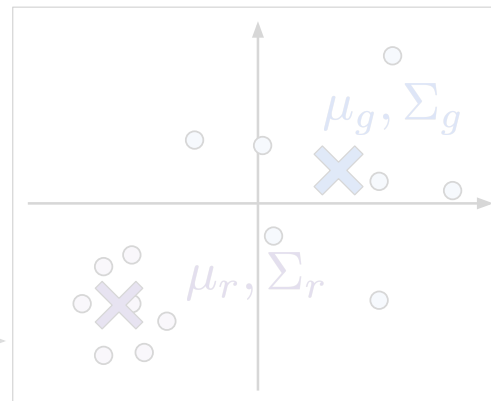


Pre-trained encoder

Generated image distribution



Pre-trained encoder



Rewinding the quarter...

Lecture 1

Lecture 2

Lecture 3

Lecture 4

Lecture 5

Lecture 6

Lecture 7

Artificial Analysis Text to Image Leaderboard Artificial Analysis

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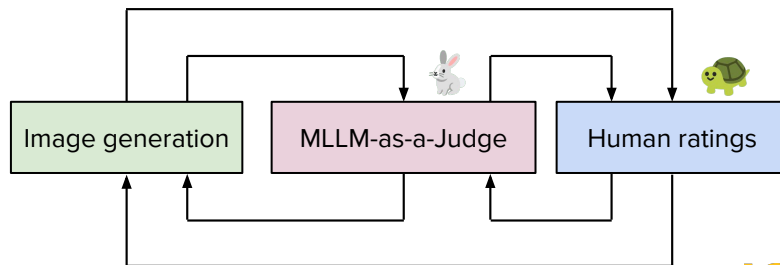
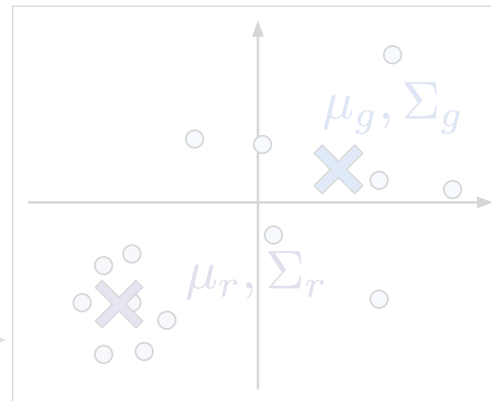


Pre-trained encoder

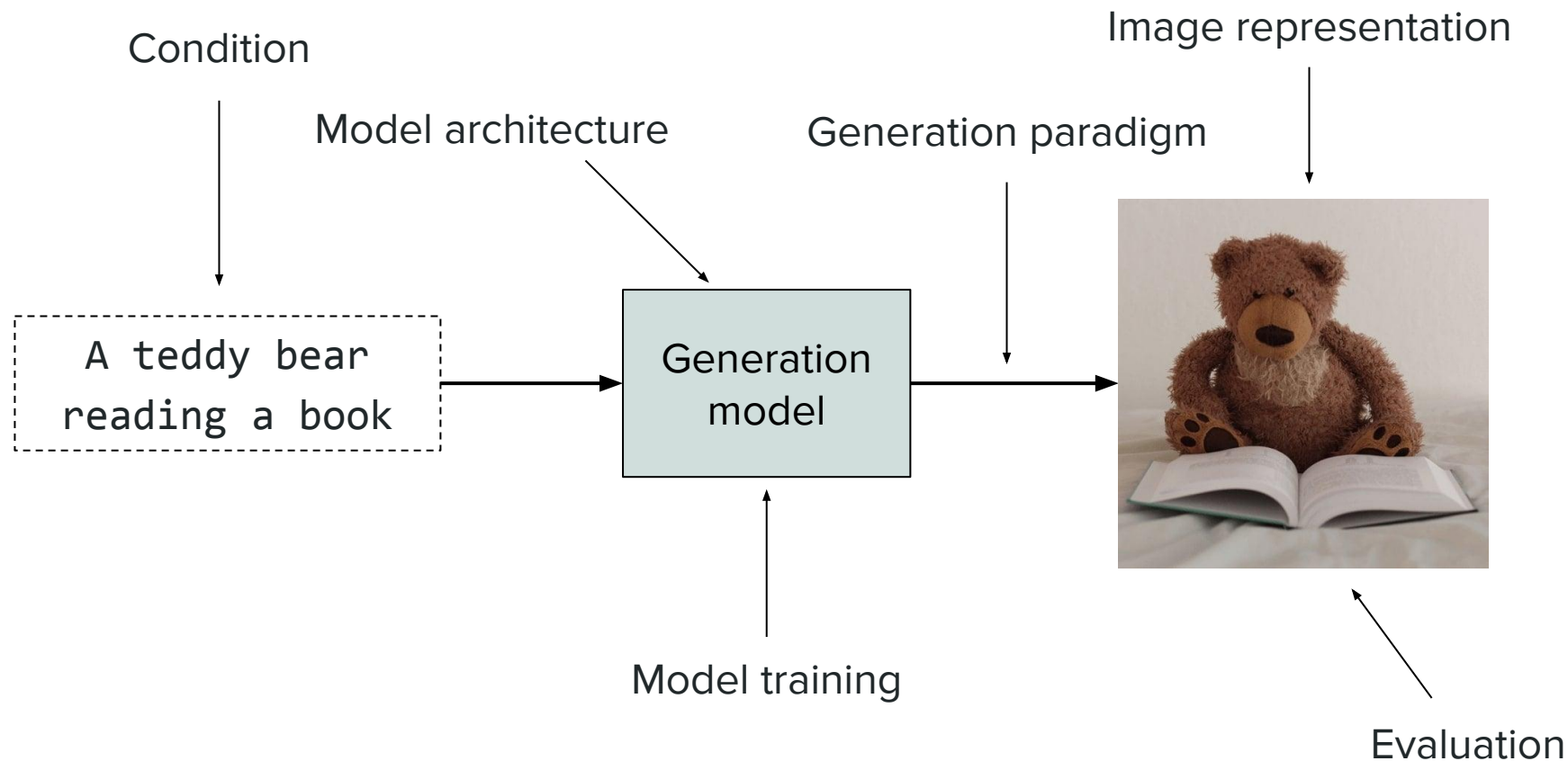
Generated image distribution



Pre-trained encoder



Rewinding the quarter...





Diffusion & Large Vision Models

Recap

State-of-the-art models

Extension to videos

Alternatives to generation

Diffusion for LLMs

Closing thoughts

Leaderboard of text-to-image models

■ Artificial Analysis Text to Image Leaderboard Artificial Analysis

↕	Range ⓘ	Creator ↕	Model ↕	Elo ↕	95% CI	Samples ↕	Released ↕	API Pricing ¹ ↕
1	1	🌀 OpenAI	GPT Image 2 (high)	1,339	-10/10	10,037	Apr 2026	\$211.0 /1k imgs
2	2-3	🌀 OpenAI	GPT Image 1.5 (high) ⓘ	1,266	-10/10	5,443	Dec 2025	\$133.0 /1k imgs
3	2-3	🌈 Google	Nano Banana 2 (Gemini 3.1 Flash Image Preview)	1,264	-10/10	7,234	Feb 2026	\$67.0 /1k imgs
4	4-5	🌈 Google	Nano Banana Pro (Gemini 3 Pro Image)	1,220	-10/10	4,712	Nov 2025	\$134.0 /1k imgs
5	4-5	🟦 xAI	grok-imagine-image-quality	1,211	-11/11	2,857	Apr 2026	\$50.0 /1k imgs

Best performing models come from **top AI labs** and are **closed-source**

Leaderboard of open weights text-to-image models

■ Artificial Analysis Text to Image Leaderboard Artificial Analysis

↕	Range ⓘ	Creator ↕	Model ↕	Elo ↕	95% CI	Samples ↕	Released ↕	API Pricing ¹ ↕
1	1	 HiDream	HiDream-O1-Image-Dev-2604  Open Weights ⓘ	1,187	−9/9	3,834	May 2026	No API available
2	2–4	 Alibaba	Qwen Image Max 2512  Open Weights	1,159	−9/9	3,992	Dec 2025	\$20.0 /1k imgs
3	2–4	 Black Forest Labs	<u>FLUX.2 [dev]</u>  Open Weights	1,159	−9/9	3,761	Nov 2025	\$12.0 /1k imgs
4	2–4	 Fal	FLUX.2 [dev] Turbo  Open Weights ⓘ	1,159	−9/9	4,378	Dec 2025	\$8.0 /1k imgs
5	5	 Fal	FLUX.2 [dev] Flash  Open Weights ⓘ	1,143	−9/9	4,361	Dec 2025	\$5.0 /1k imgs

...but **open-weights models** are **not too far** behind!

Leaderboard of open weights text-to-image models

1	1	 HiDream	HiDream-O1-Image-Dev-2604  Open Weights ⓘ	1,187	−9/9	3,834	May 2026	No API available
2	2–4	 Alibaba	Qwen Image Max 2512  Open Weights	1,159	−9/9	3,992	Dec 2025	\$20.0 /1k imgs
3	2–4	 Black Forest Labs	<u>FLUX.2 [dev]</u>  Open Weights	1,159	−9/9	3,761	Nov 2025	\$12.0 /1k imgs
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5	5	 Fal	FLUX.2 [dev] Flash  Open Weights ⓘ	1,143	−9/9	4,361	Dec 2025	\$5.0 /1k imgs

The usual.

- Rectified **flow**
 - Relies on **VAE**
 - Hybrid dual/single-stream
 - Uses **Mistral 3** embeddings
- MM-DiT**

Leaderboard of **open weights** text-to-image models

1	1	 HiDream	HiDream-O1-Image-Dev-2604  	1,187	−9/9	3,834	May 2026	No API available
2	2–4	 Alibaba	Qwen Image Max 2512 	1,159	−9/9	3,992	Dec 2025	\$20.0 /1k imgs
3	2–4	 Black Forest Labs	FLUX.2 [dev] 	1,159	−9/9	3,761	Nov 2025	\$12.0 /1k imgs
4	2–4	 Fal	FLUX.2 [dev] Turbo  	1,159	−9/9	4,378	Dec 2025	\$8.0 /1k imgs
5	5	 Fal	FLUX.2 [dev] Flash  	1,143	−9/9	4,361	Dec 2025	\$5.0 /1k imgs

The usual.

- **Flow matching**-based loss
- Relies on **VAE**
- Double-stream **MM-DiT**
- Uses **Qwen2.5VL** embeddings

Trend to watch: back to pixel-space diffusion?

1	1	 HiDream	HiDream-O1-Image-Dev-2604  Open Weights ⓘ	1,187	−9/9	3,834	May 2026	No API available
2	2–4	 Alibaba	Qwen Image Max 2512  Open Weights	1,159	−9/9	3,992	Dec 2025	\$20.0 /1k imgs
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5	5	 Fal	FLUX.2 [dev] Flash  Open Weights ⓘ	1,143	−9/9	4,361	Dec 2025	\$5.0 /1k imgs

The usual.

- Flow matching-based loss
- Transformer-based model

What stands out!

- VAE removed
- No disjoint pre-trained text encoder



Diffusion & Large Vision Models

Recap

State-of-the-art models

Extension to videos

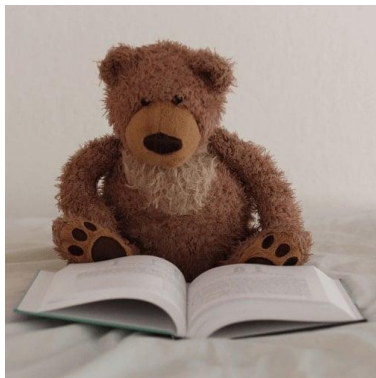
Alternatives to generation

Diffusion for LLMs

Closing thoughts

From image to video generation

- Extra dimension: **time**



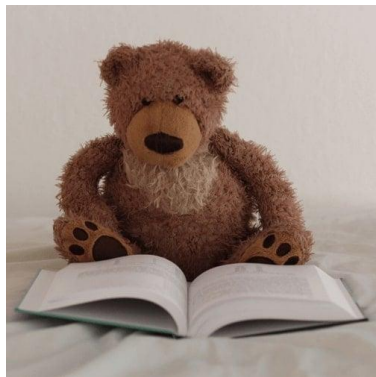
Image



Video

From image to video generation

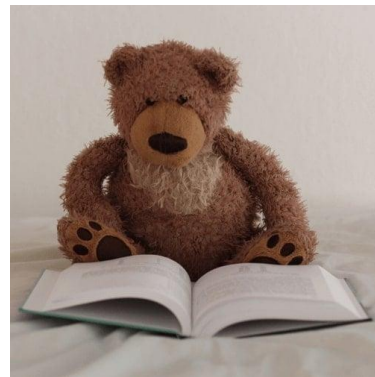
- Extra dimension: **time**
- Temporal **consistency** from one frame to another



t



$t + 1$



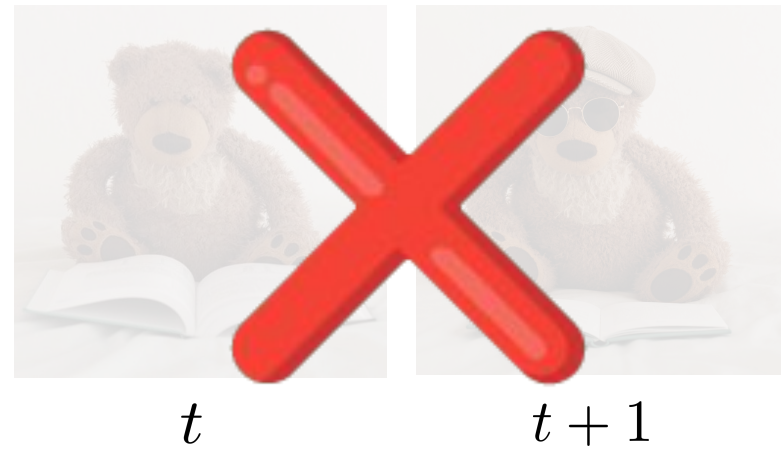
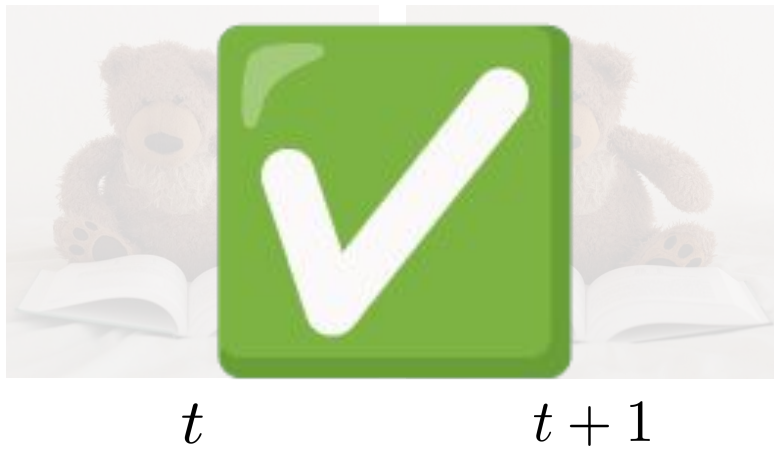
t



$t + 1$

From image to video generation

- Extra dimension: **time**
- Temporal **consistency** from one frame to another



From image to video generation

- Extra dimension: **time**
- Temporal **consistency** from one frame to another
- **Computation** considerations

height $\times$ width $\times$ time
can be **large**

From image to video generation

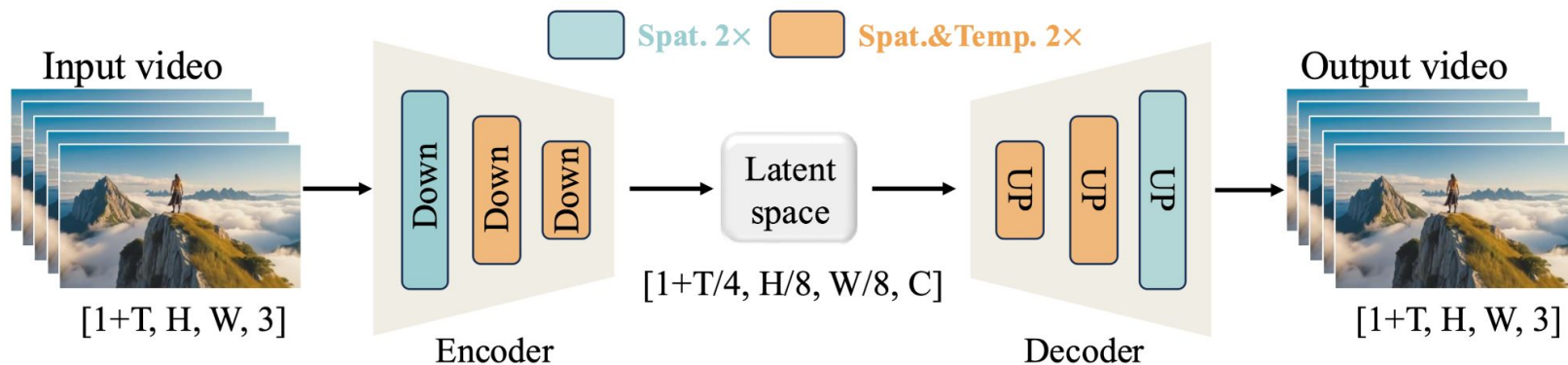
- Extra dimension: **time**
- Temporal **consistency** from one frame to another
- **Computation** considerations
- **Metrics**

$$\text{FVD} = \|\mu_r - \mu_g\|^2 + \text{Tr} \left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2} \right)$$

Fréchet **V**ideo **D**istance

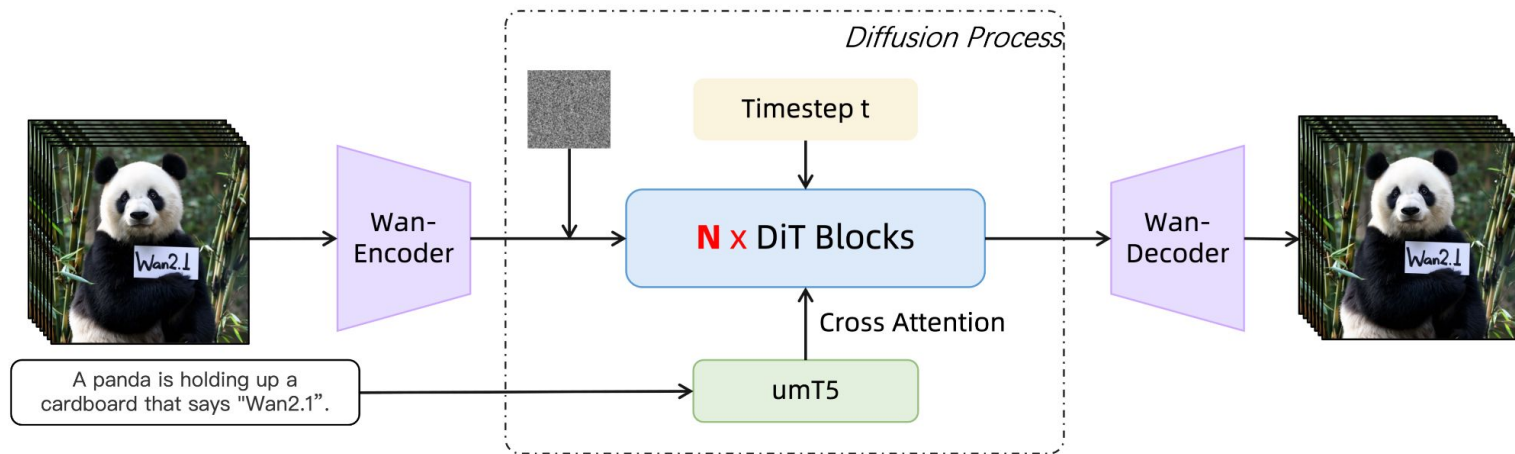
Exploring architecture of leading model

- **Spatio-temporal VAE**



Exploring architecture of leading model

- Spatio-temporal VAE
- Model **architecture**



Video generation suggested readings

- Video Diffusion Models, Ho et al., 2022
- MagicVideo: Efficient Video Generation with Latent Diffusion Models, Zhou et al., 2022
- Stable Video Diffusion: Scaling Latent Video Diffusion Models to Large Datasets, Blattmann et al., 2023
- Video generation models as world simulators, OpenAI, 2024
- Movie Gen: A Cast of Media Foundation Models, Meta Team, 2024
- HunyuanVideo: A Systematic Framework For Large Video Generative Models, Kong et al., 2024
- Wan: Open and Advanced Large-Scale Video Generative Models, Team Wan, 2025

Note: this list is **not exhaustive**.



Diffusion & Large Vision Models

Recap

State-of-the-art models

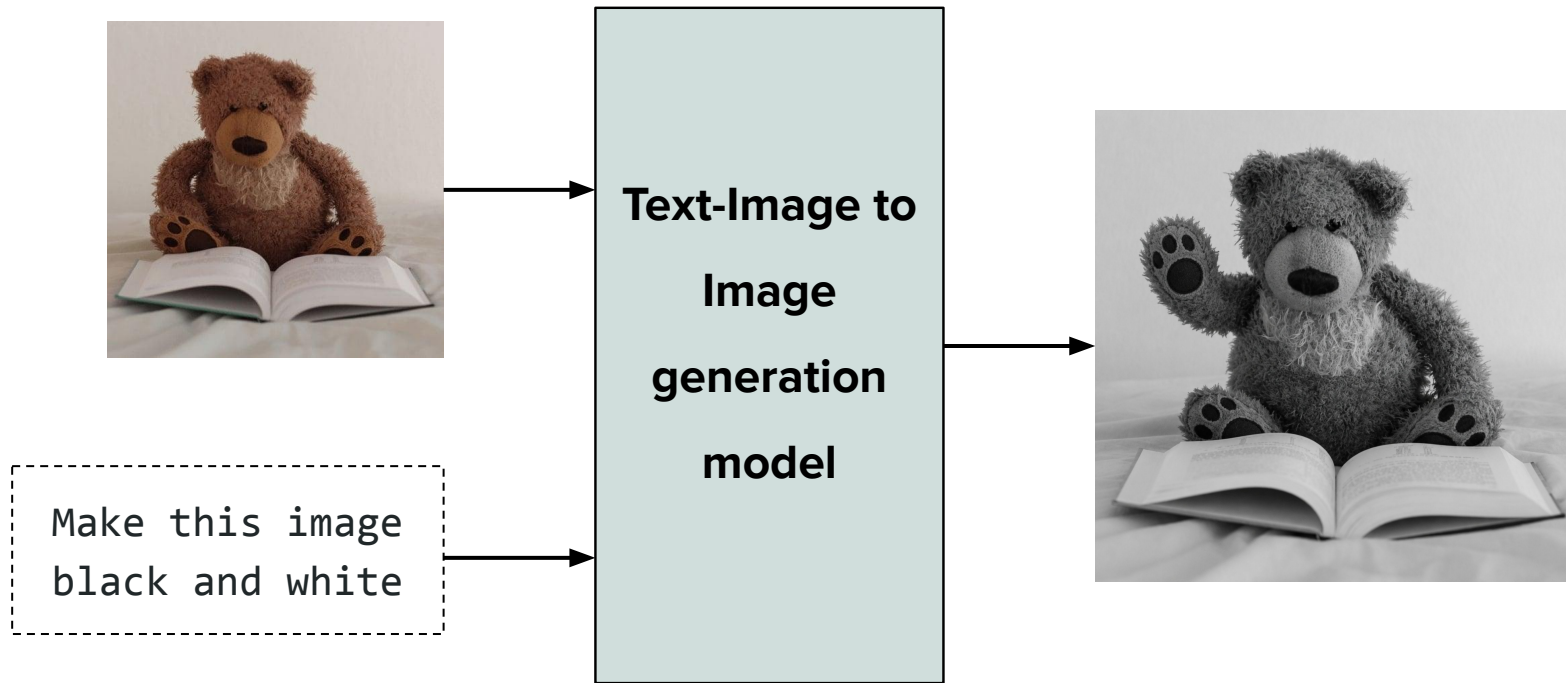
Extension to videos

Alternatives to generation

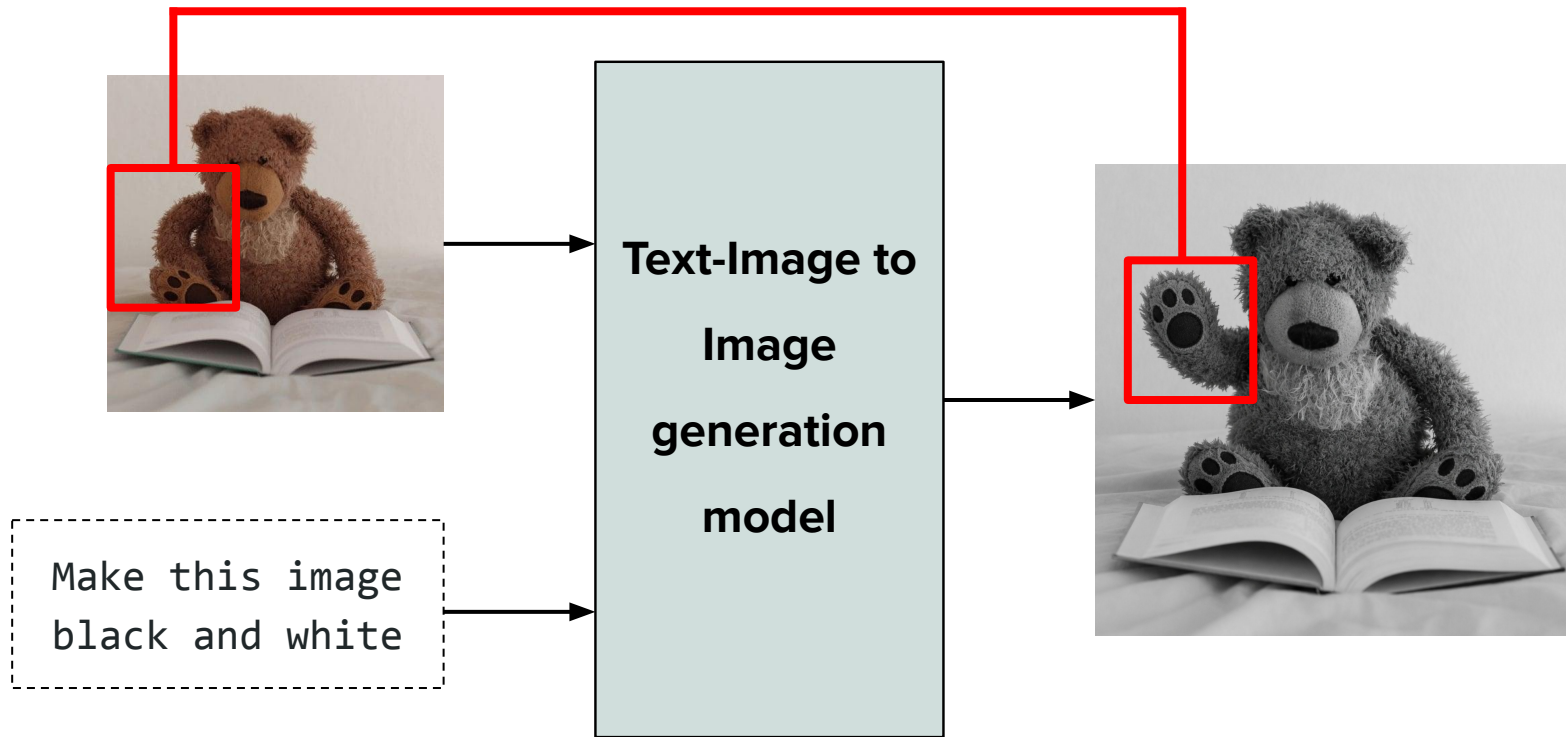
Diffusion for LLMs

Closing thoughts

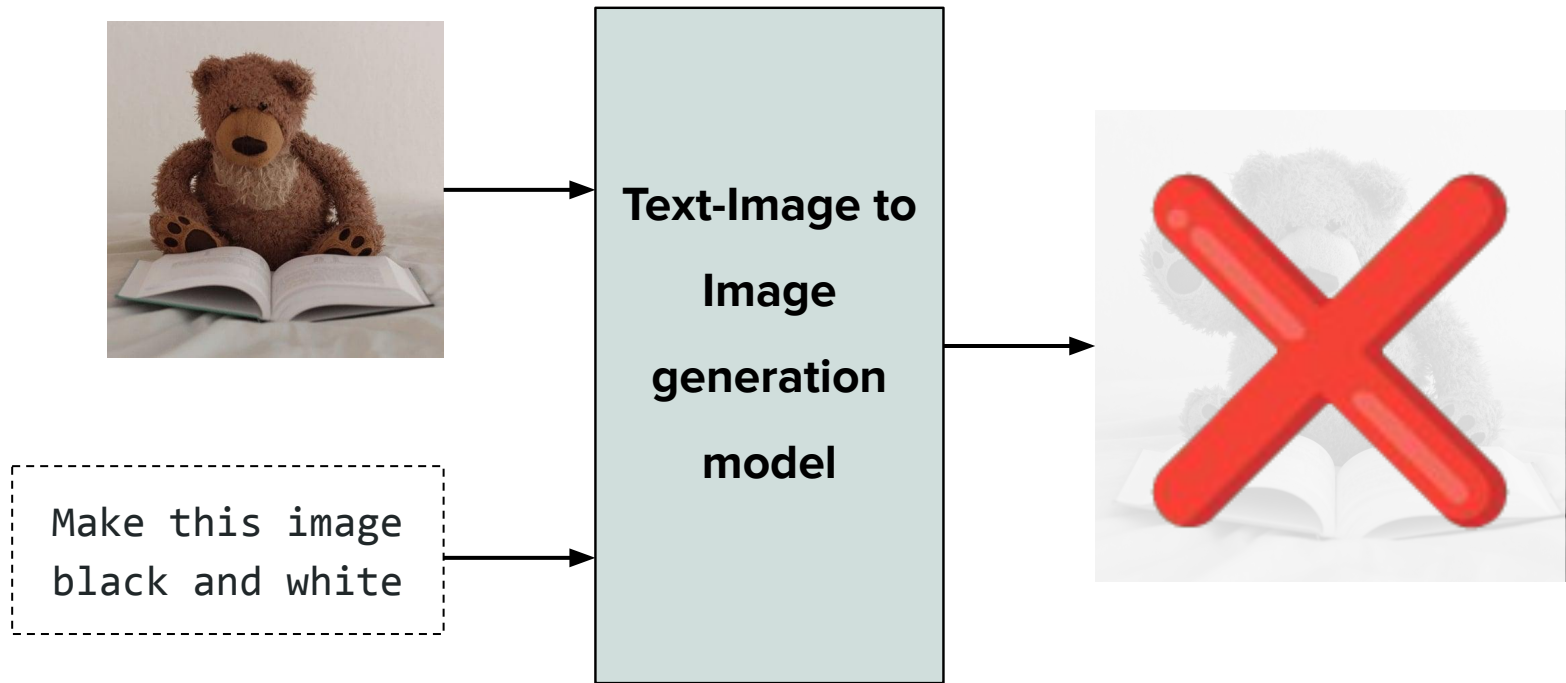
Limitation of image generation models



Limitation of image generation models



Limitation of image generation models



Reformulate problem to put proper guardrails

Idea. Avoid from-scratch generation that is prone to mistakes by putting guardrails

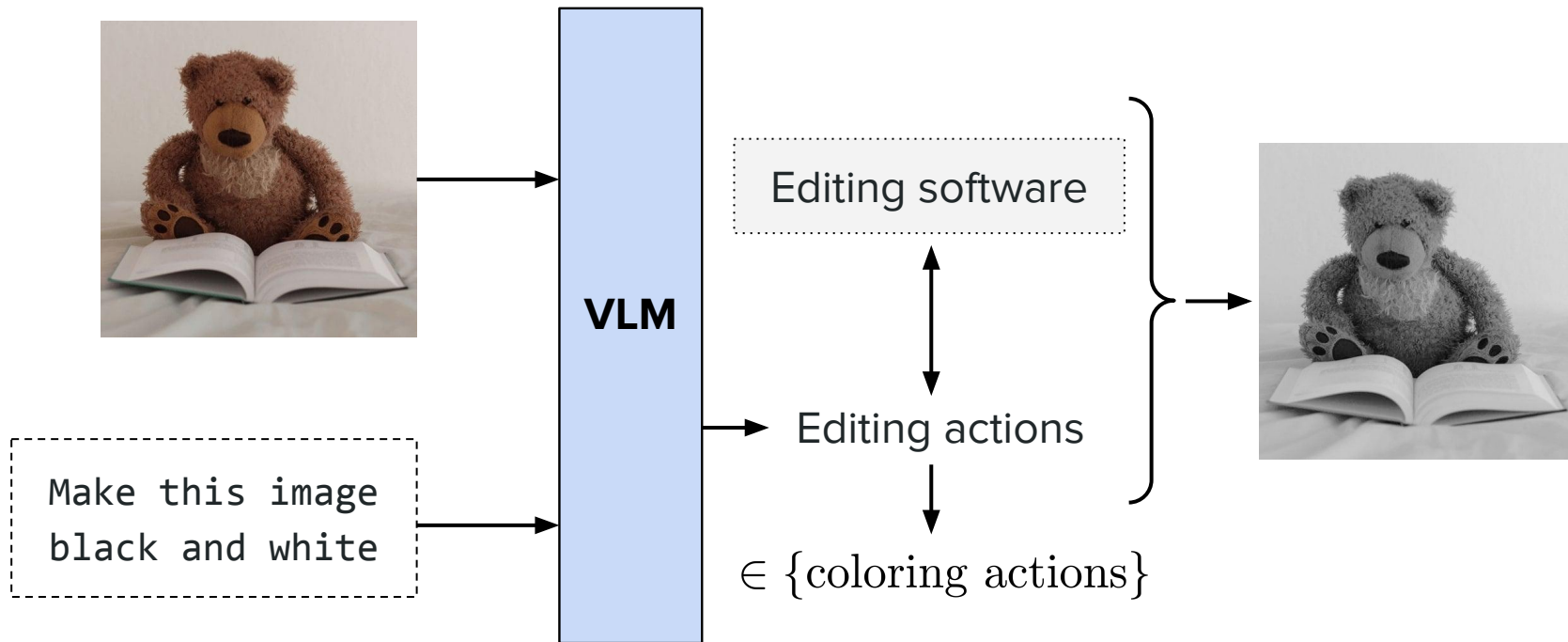


Image editing suggested readings

- SmartEdit: Exploring Complex Instruction-based Image Editing with Multimodal Large Language Models, Huang et al., 2024
- MonetGPT: Solving Puzzles Enhances MLLMs' Image Retouching Skills, Dutt et al., 2025
- JarvisArt: Liberating Human Artistic Creativity via an Intelligent Photo Retouching Agent, Lin et al., 2025
- RetouchIQ: MLLM Agents for Instruction-Based Image Retouching with Generalist Reward, Wu et al., 2026

Note: this list is **not exhaustive**.



Diffusion & Large Vision Models

Recap

State-of-the-art models

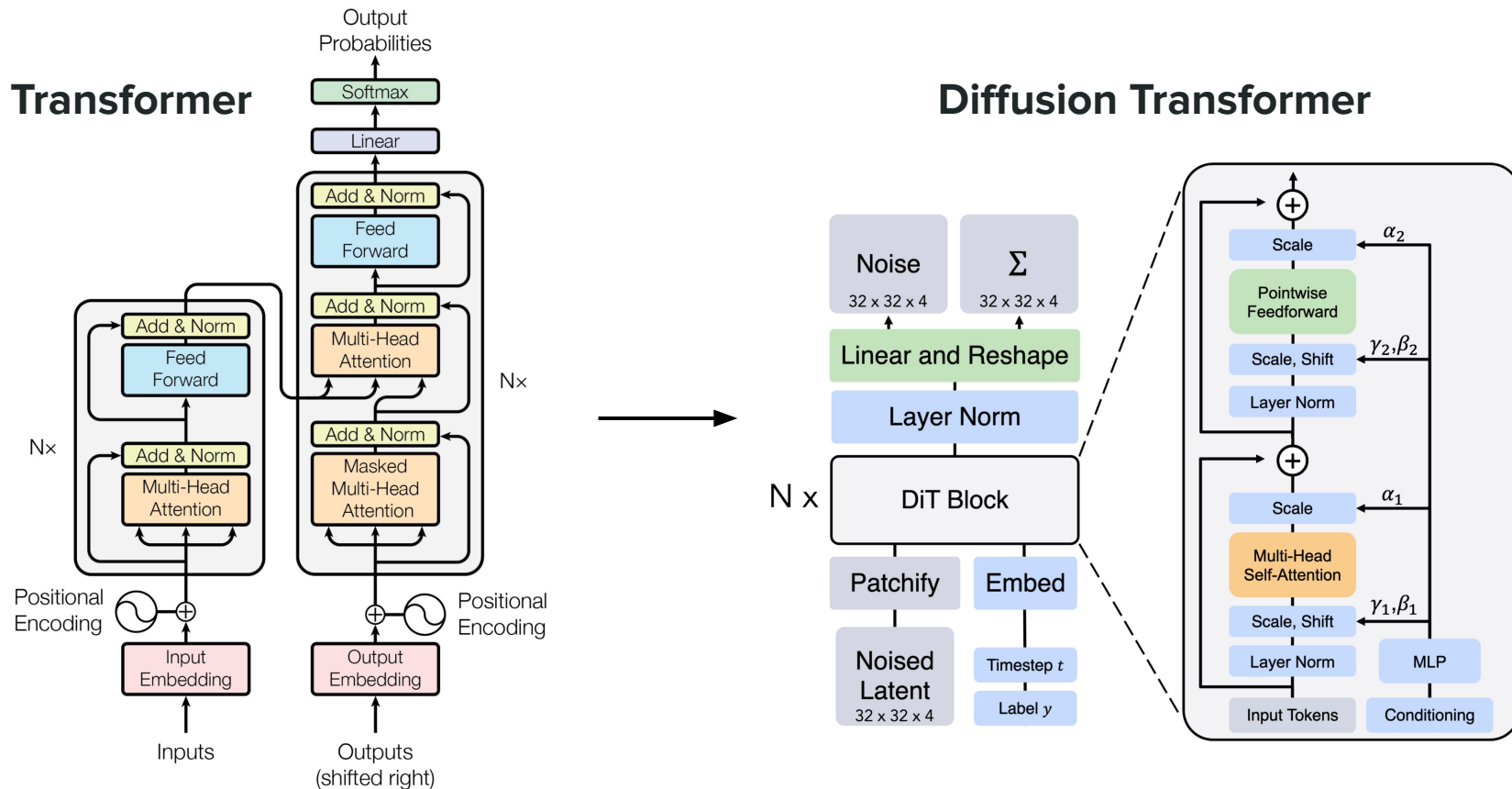
Extension to videos

Alternatives to generation

Diffusion for LLMs

Closing thoughts

Concepts borrowed from the text world



Concepts borrowed from the text world

DPO

**Direct Preference Optimization:
Your Language Model is Secretly a Reward Model**

Rafael Rafailov^{*†}

Archit Sharma^{*†}

Eric Mitchell^{*†}

Stefano Ermon^{†‡}

Christopher D. Manning[†]

Chelsea Finn[†]

Diffusion-DPO

Diffusion Model Alignment Using Direct Preference Optimization

Bram Wallace¹

Meihua Dang²

Rafael Rafailov²

Linqi Zhou²

Aaron Lou²

Senthil Purushwalkam¹

Stefano Ermon²

Caiming Xiong¹

Shafiq Joty¹

Nikhil Naik¹

Concepts borrowed from the text world

DPO

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Diffusion-DPO

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Nikhil Naik¹

GRPO

DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models

Zhihong Shao^{1,2,*†}, Peiyi Wang^{1,3,*†}, Qihao Zhu^{1,3,*†}, Runxin Xu¹, Junxiao Song¹
Xiao Bi¹, Haowei Zhang¹, Mingchuan Zhang¹, Y.K. Li¹, Y. Wu¹, Daya Guo^{1*}



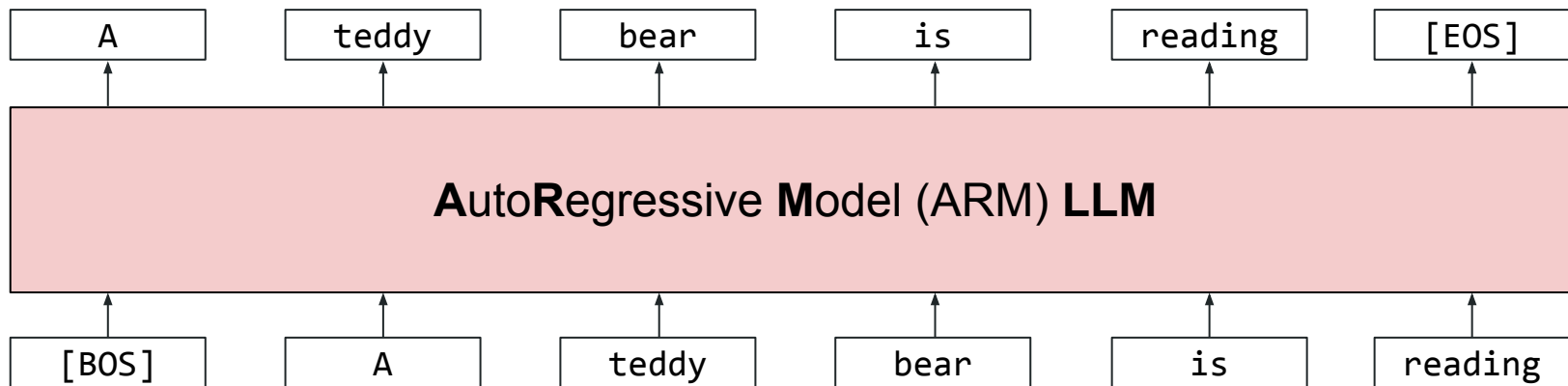
Flow-GRPO

Flow-GRPO:
Training Flow Matching Models via Online RL

Jie Liu^{1,3,5,*} Gongye Liu^{2,3,*} Jiajun Liang³ Yangguang Li¹
Jiaheng Liu⁴ Xintao Wang³ Pengfei Wan³ Di Zhang³ Wanli Ouyang^{1,5†}

Traditional generation paradigm for text

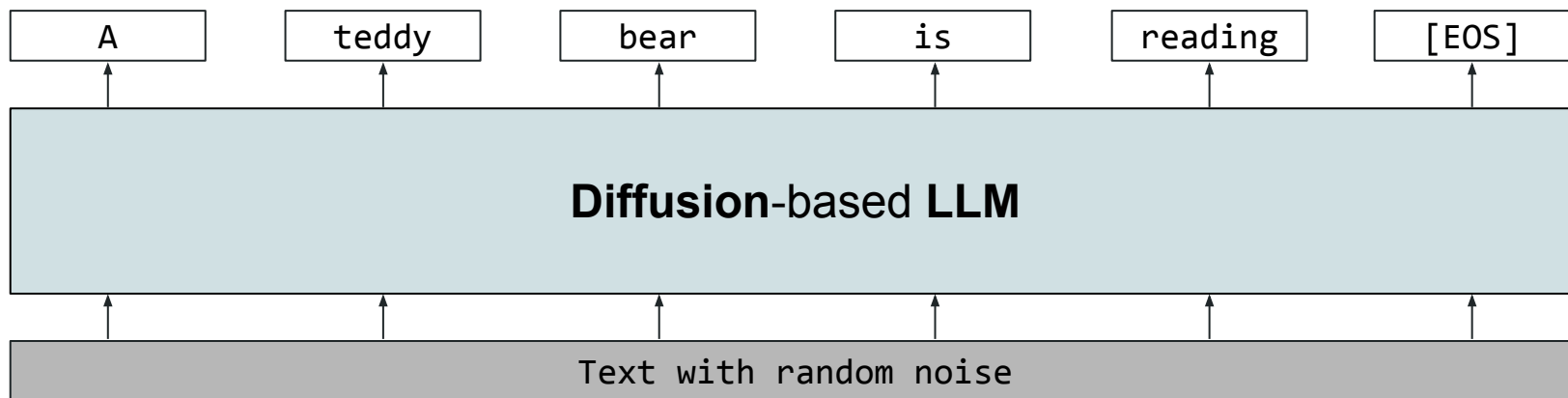
Context. Text generation models generate rely on "next token prediction"



Problem. Inference-time generation needs $\mathcal{O}(\# \text{ output tokens})$ iterations

What about diffusion for text?

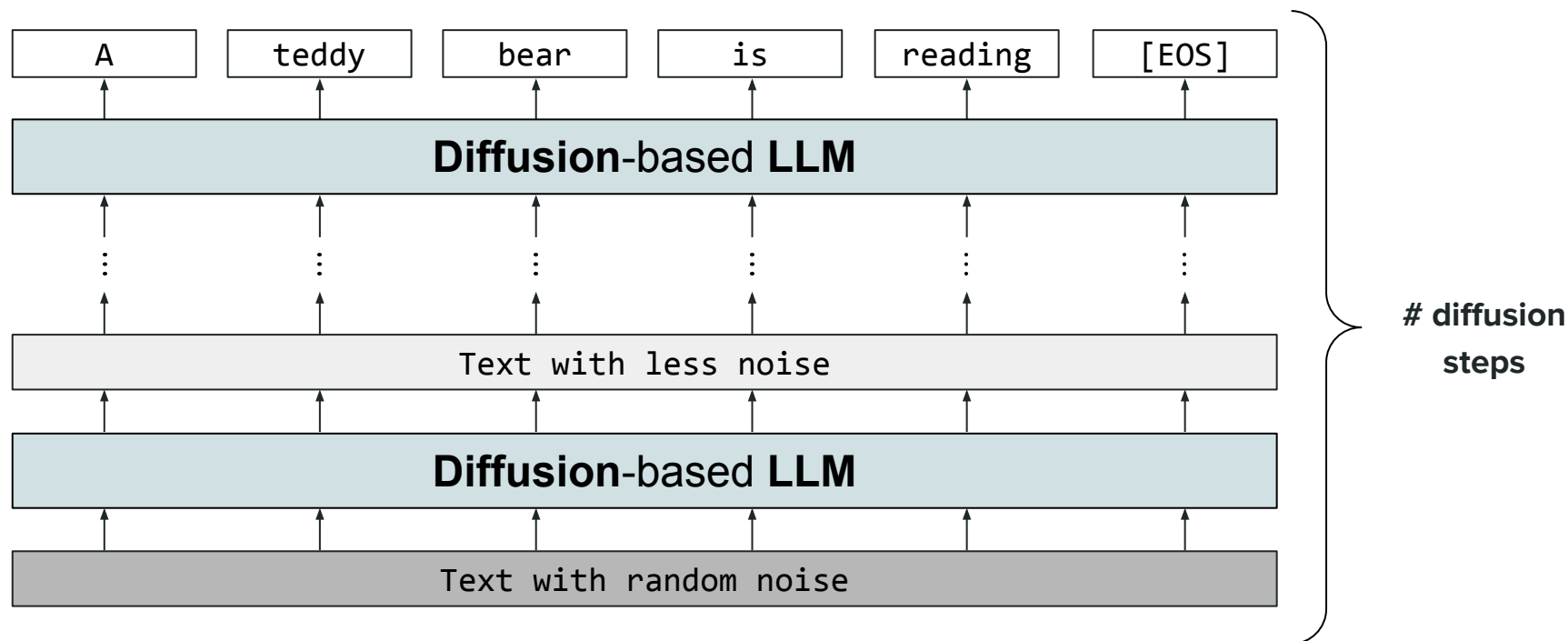
Idea. Adapt diffusion paradigm to text generation problem



Move number of iterations from $\mathcal{O}(\# \text{ output tokens})$ to $\mathcal{O}(\# \text{ diffusion steps})$

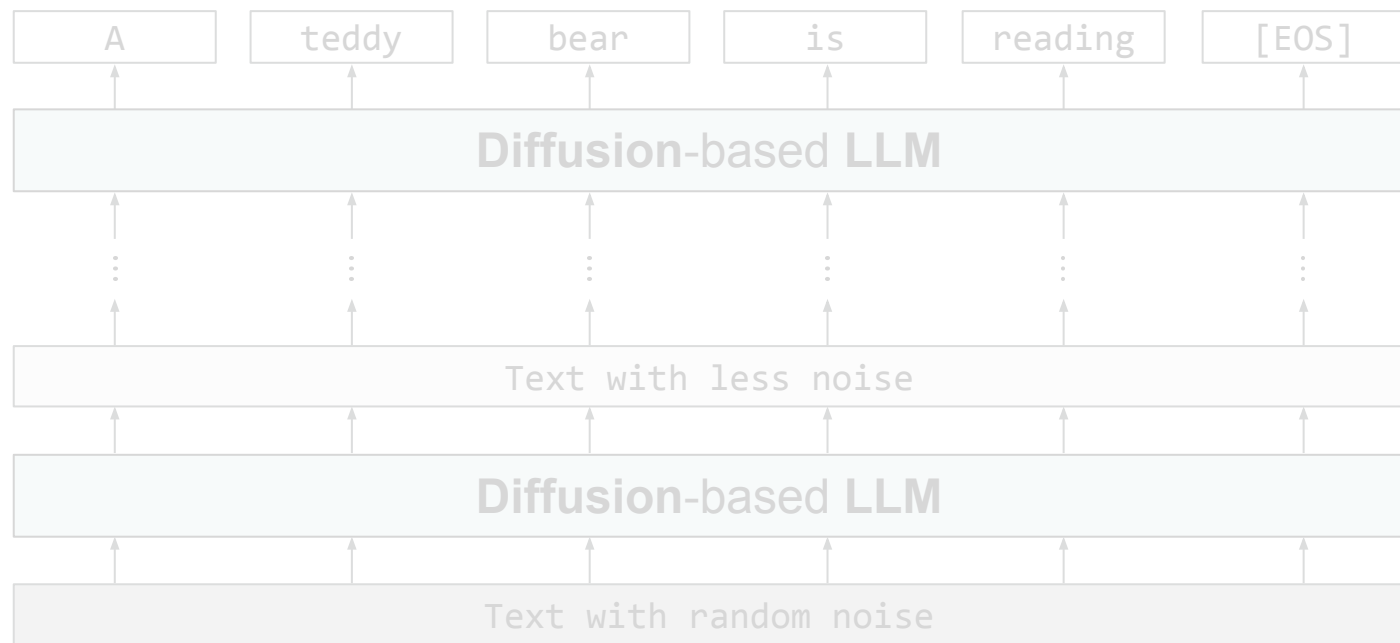
What about diffusion for text?

Idea. Adapt diffusion paradigm to text generation problem



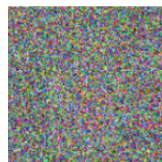
What about diffusion for text?

Idea. Adapt diffusion paradigm to text generation problem



Finding the equivalent of noise for text

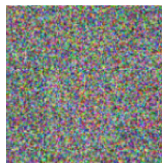
Noisy image



Noisy text

Finding the equivalent of noise for text

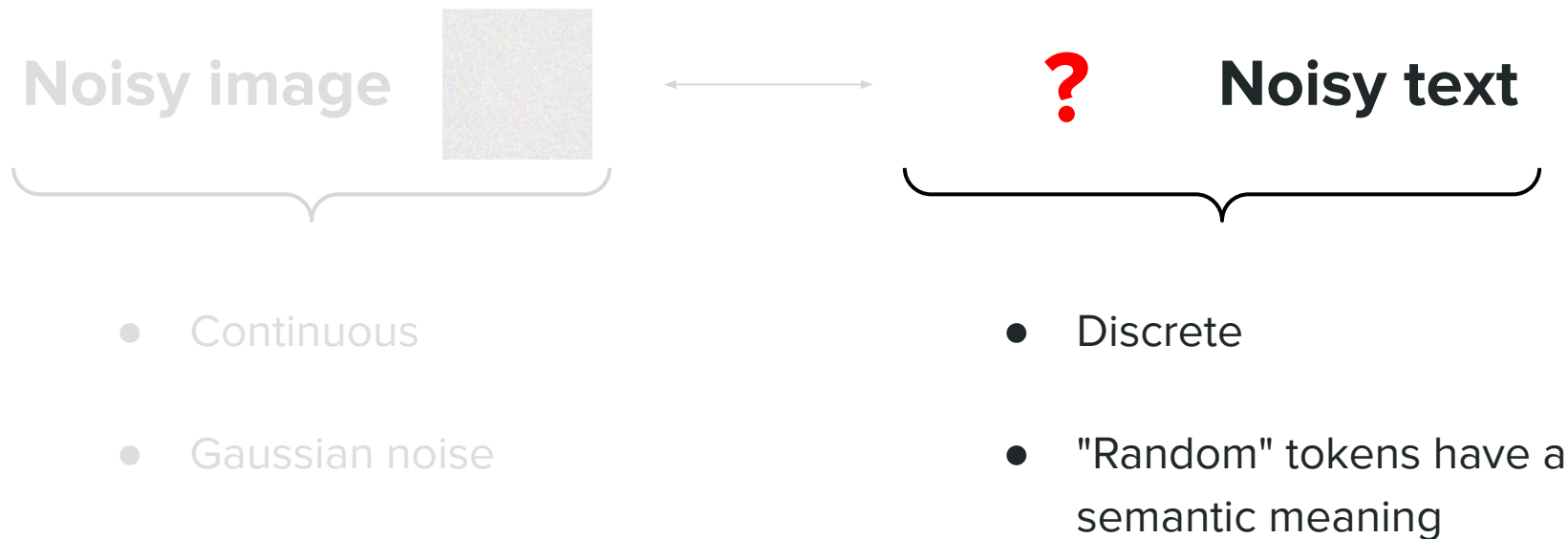
Noisy image



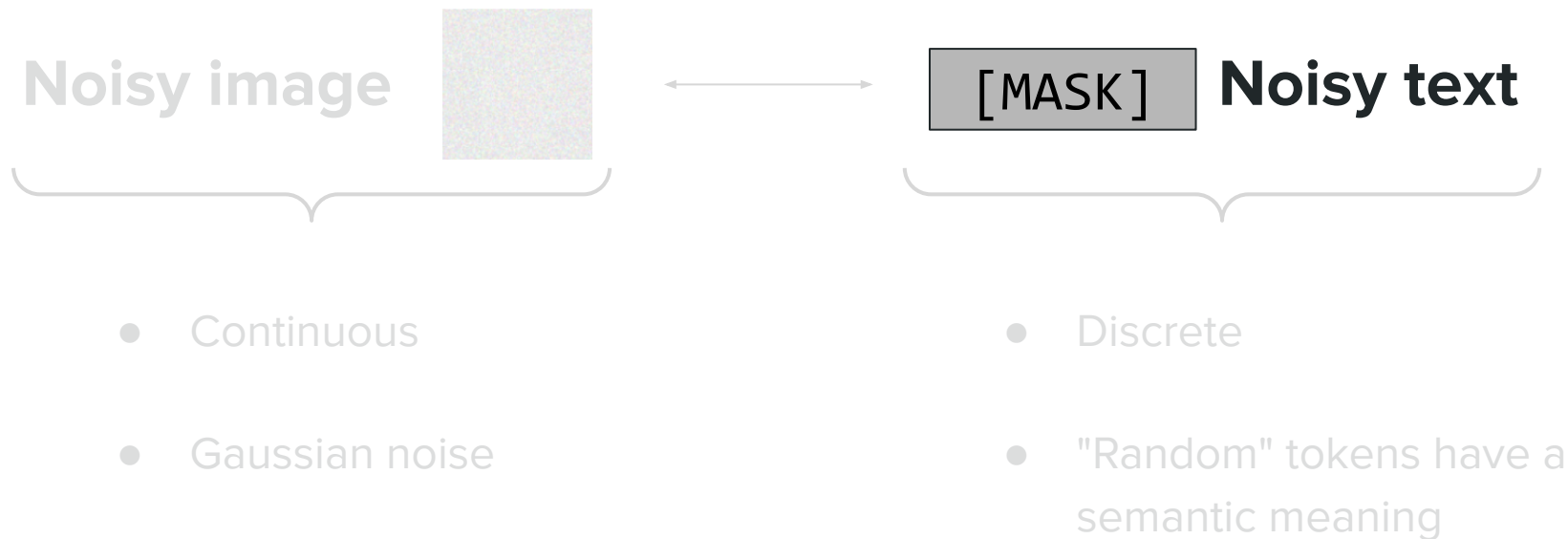
Noisy text

- Continuous
- Gaussian noise

Finding the equivalent of noise for text

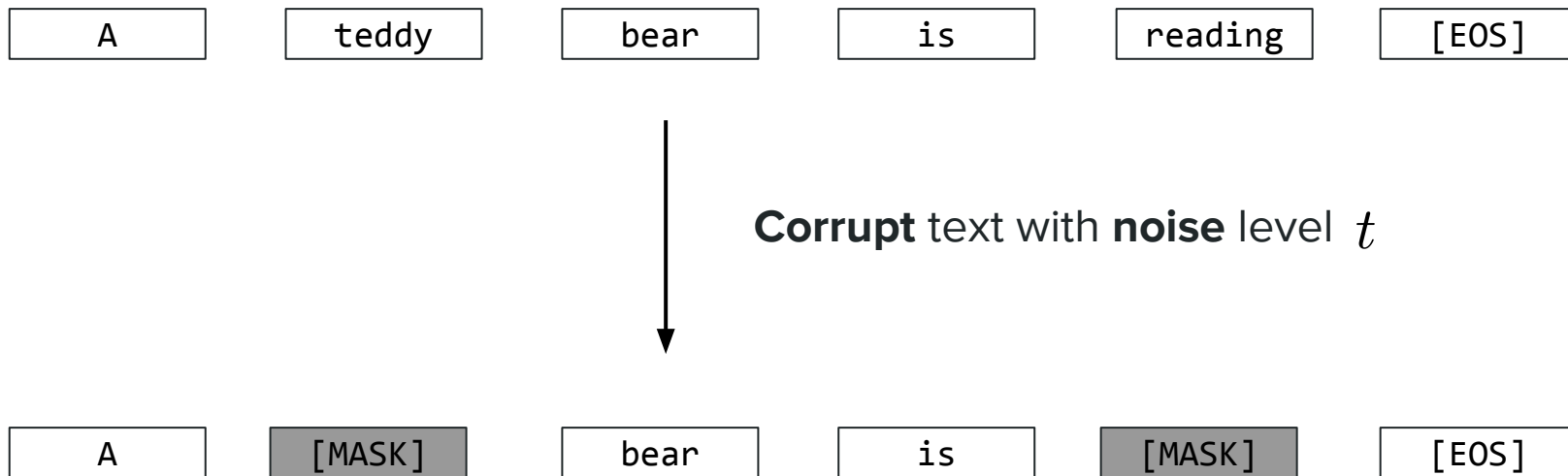


Finding the equivalent of noise for text



Training: learn to fill the blanks

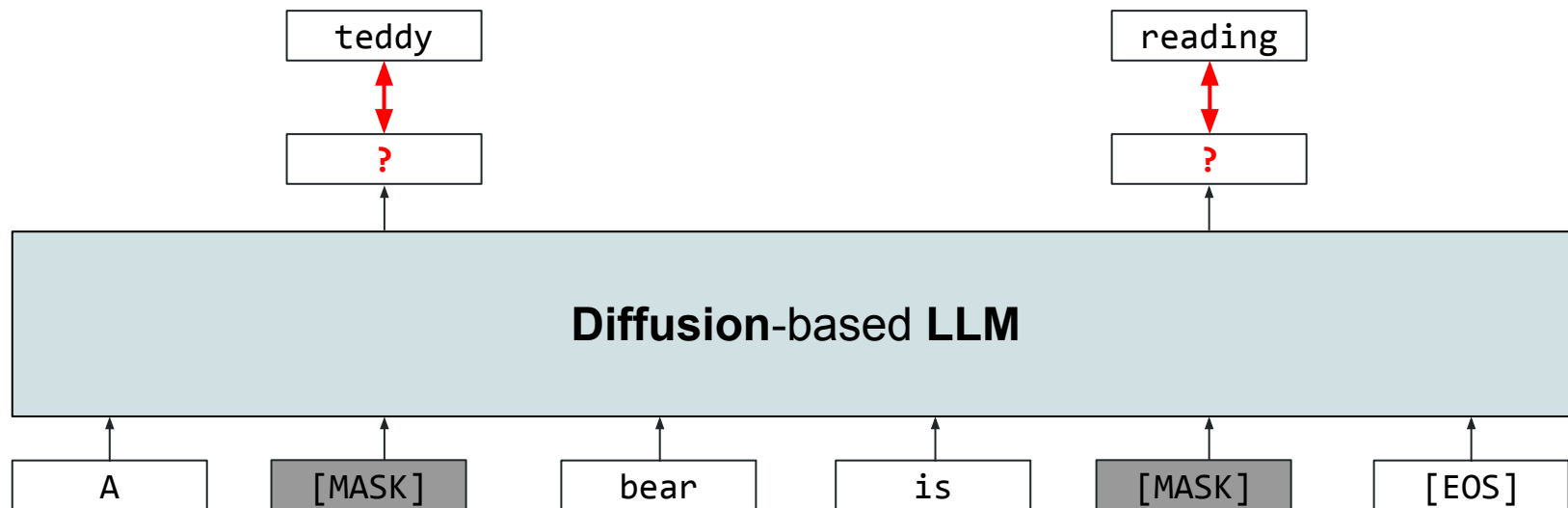
Step 1. Sample noise $t \sim \mathcal{U}(0, 1)$ and mask ratio t of tokens



Training: learn to fill the blanks

Step 1. Sample noise $t \sim \mathcal{U}(0, 1)$ and mask ratio t of tokens

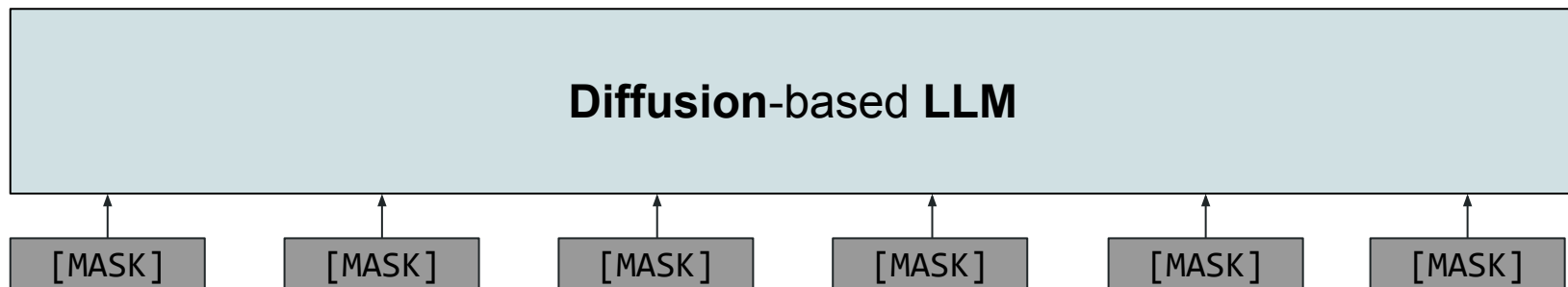
Step 2. Predict tokens that were masked



Inference: coarse to fine

Initialize. Start with masked input

$t = 100\%$ noise

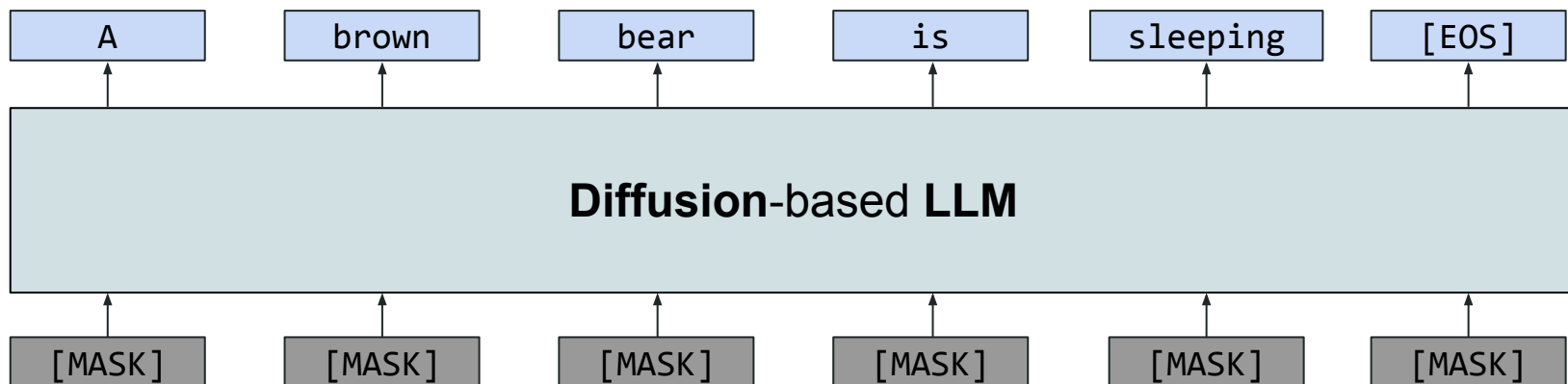


Inference: coarse to fine

Initialize. Start with masked input

$t = 100\%$ noise

Iterate. Predict masked tokens (**commit**)

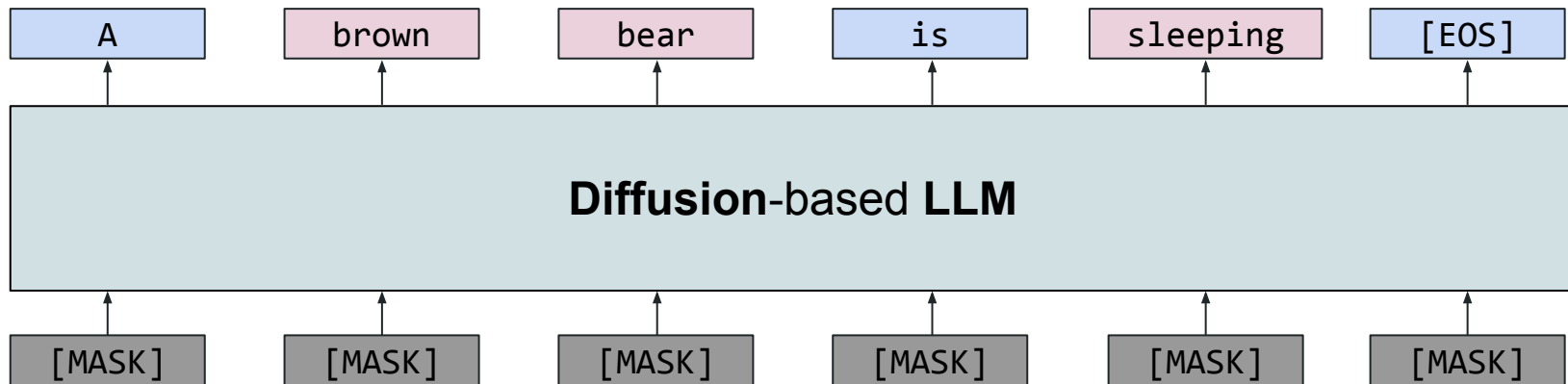


Inference: coarse to fine

Initialize. Start with masked input

$t = 100\%$ noise

Iterate. Predict masked tokens (**commit**) and perform remasking (**revise**)

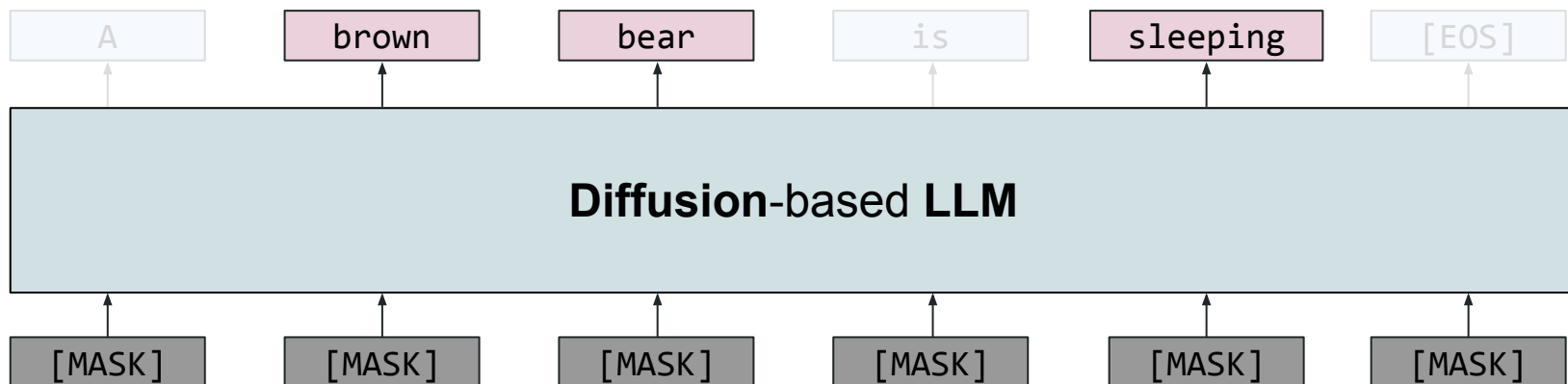


Inference: coarse to fine

Initialize. Start with masked input

$t = 100\%$ noise

Iterate. Predict masked tokens (**commit**) and perform remasking (**revise**)

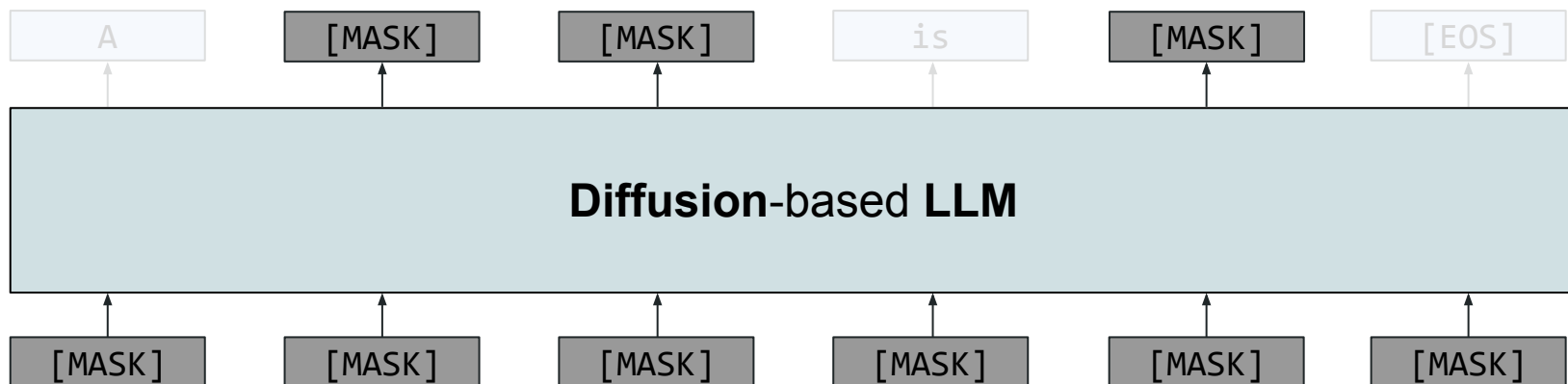


Inference: coarse to fine

Initialize. Start with masked input

$t = 100\%$ noise

Iterate. Predict masked tokens (**commit**) and perform remasking (**revise**)

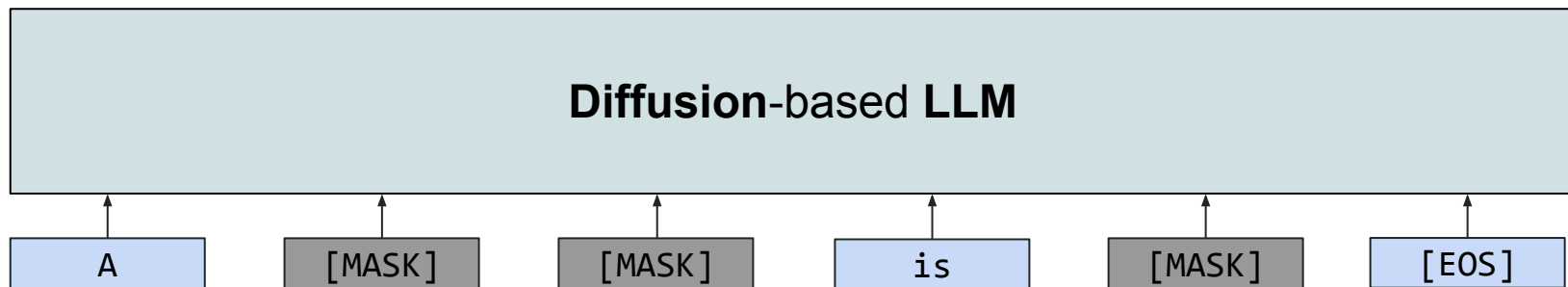


Inference: coarse to fine

Initialize. Start with masked input

$t = 50\%$ noise

Iterate. Predict masked tokens (**commit**) and perform remasking (**revise**)



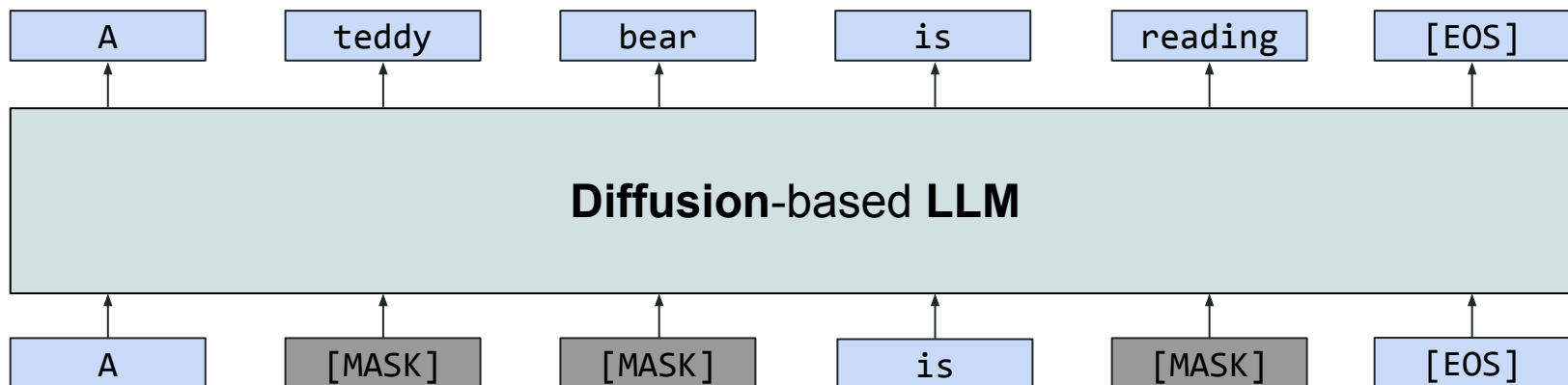
Inference: coarse to fine

Initialize. Start with masked input

$t = 50\%$ noise

Iterate. Predict masked tokens (**commit**) and perform remasking (**revise**)

Output. Obtain final output



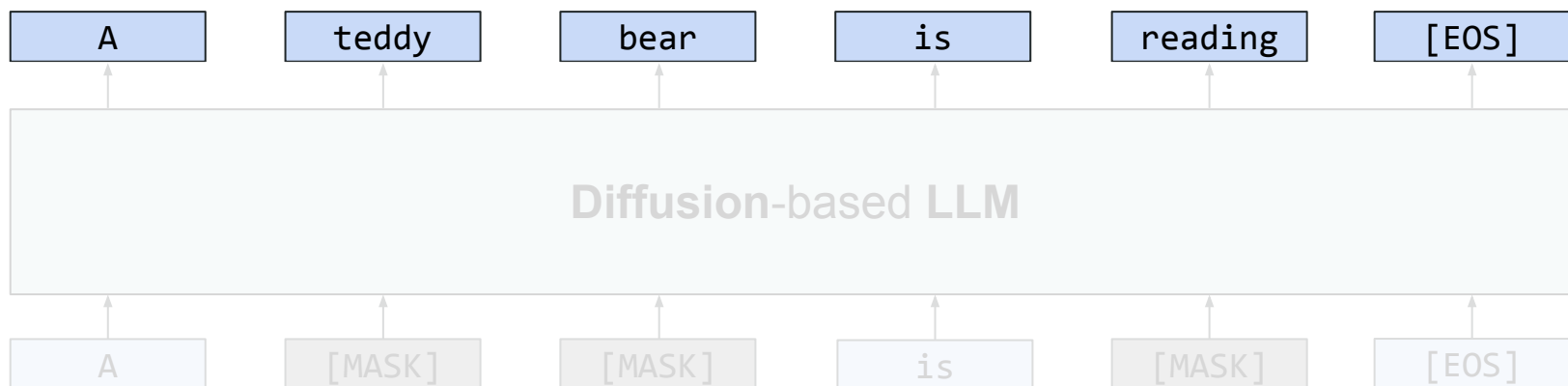
Inference: coarse to fine

Initialize. Start with masked input

$t = 0\%$ noise

Iterate. Predict masked tokens (**commit**) and perform remasking (**revise**)

Output. Obtain final output



Discussion

Advantages.

-  **10x speed** compared to ARM

Discussion

Advantages.

- ⚡ 10x speed compared to ARM
- 💻 Well suited for **coding** tasks
 - ➔ **"Fill in the middle"** tasks
 - ➔ **Speed** is a **bottleneck** for "Coding agents"

Discussion

Advantages.

- ⚡ **10x speed** compared to ARM
- 💻 Well suited for **coding** tasks

"Fill in the middle" tasks

Speed is a **bottleneck** for "Coding agents"

Challenges and current work.

- 🏋️ **Training** is significantly more **expensive**

Discussion

Advantages.

- ⚡ **10x speed** compared to ARM
 - 💻 Well suited for **coding** tasks
- Two arrows originate from the right side of the second bullet point, pointing to the following text:
- "Fill in the middle" tasks
 - **Speed** is a **bottleneck** for "Coding agents"

Challenges and current work.

- 🏋️ **Training** is significantly more **expensive**
- 🔧 **Adapt** ARM-based techniques (e.g. post-training approaches)

Discussion

Advantages.

- ⚡ **10x speed** compared to ARM
 - 💻 Well suited for **coding** tasks
- "Fill in the middle" tasks
- **Speed** is a **bottleneck** for "Coding agents"

Challenges and current work.

- 🏋️ **Training** is significantly more **expensive**
- 🛠️ **Adapt** ARM-based techniques (e.g. post-training approaches)
- 🤝 **Combine** benefits of **autoregression** and **diffusion**

Diffusion LLMs is currently a very hot topic!



Beyond GPT architecture: Why Google's Diffusion approach could reshape LLM deployment

VentureBeat article, June 13th, 2025.

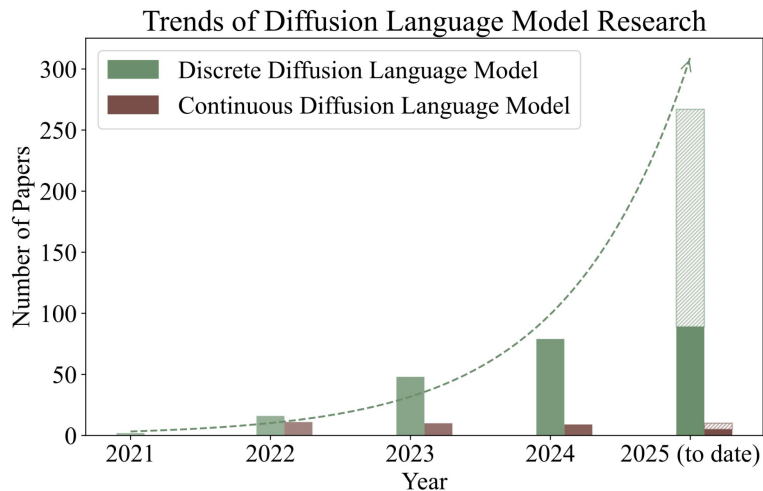


Figure from A Survey on Diffusion Language Models, Li et al., 2025.

Inception raises \$50 million to build diffusion models for code and text

TechCrunch headline, November 6th 2025.



[Newsroom](#) [Services & Solutions](#) [Resources](#) [For Journalists](#)

Feb 24, 2026 11:00 AM Eastern Standard Time

Inception Launches Mercury 2, the Fastest Reasoning LLM — 5x Faster Than Leading Speed-Optimized LLMs, with Dramatically Lower Inference Cost

BusinessWire article, February 24th, 2026.

Diffusion LLMs suggested readings

- Structured Denoising Diffusion Models in Discrete State-Spaces, Austin et al., 2021
- Scaling up Masked Diffusion Models on Text, Nie et al., 2024
- Large Language Diffusion Models, Nie et al., 2025
- Block Diffusion: Interpolating Between Autoregressive and Diffusion Language Models, Arriola et al., 2025
- Seed Diffusion: A Large-Scale Diffusion Language Model with High-Speed Inference, Song et al., 2025
- LLaDA2.0: Scaling Up Diffusion Language Models to 100B, Bie et al., 2025
- dLLM: Simple Diffusion Language Modeling, Zhou et al., 2026

Note: this list is **not exhaustive**.



Diffusion & Large Vision Models

Recap

State-of-the-art models

Extension to videos

Alternatives to generation

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Closing thoughts

More use cases to be expected

Tomorrow.

Democratization of existing generation capabilities.

** Image output is priced at \$120 per 1,000,000 tokens. Output images from 1024x1024px (1K) and up to 2048x2048px (2K) consume 1120 tokens and are equivalent to \$0.134 per image. Output images up to 4096x4096px (4K) consume 2000 tokens and are equivalent to \$0.24 per image.

Gemini 3 Pro Image Preview

Generated image: The first generated megapixel is charged \$0.07. Each subsequent megapixel is charged \$0.03.

FLUX.2 [max]

GPT Image 2

Quality	1024 x 1024
Low	\$0.006
Medium	\$0.053
High	\$0.211

The Ultra service uses 8 credits per successful result. You will not be charged for failed results.

Stable Image Ultra

More use cases to be expected

Tomorrow.

Democratization of existing generation capabilities.

Near term.

Integrate wins seen across modalities.

Reasoning

Truly insightful diagram?

Coding

Create image using Photoshop/CAD?

Deep Research

Synthesize entire class?

More use cases to be expected

Tomorrow.

Democratization of existing generation capabilities.

Near term.

Integrate wins seen across modalities.

Long term.

Translation of consequences across industries.

Robotics

Custom environments?

Medicine

Accelerated MRIs?

Visual design

Faster cycles?

More use cases to be expected

Tomorrow.

Democratization of existing generation capabilities.

Near term.

Integrate wins seen across modalities.

Long term.

Translation of consequences across industries.

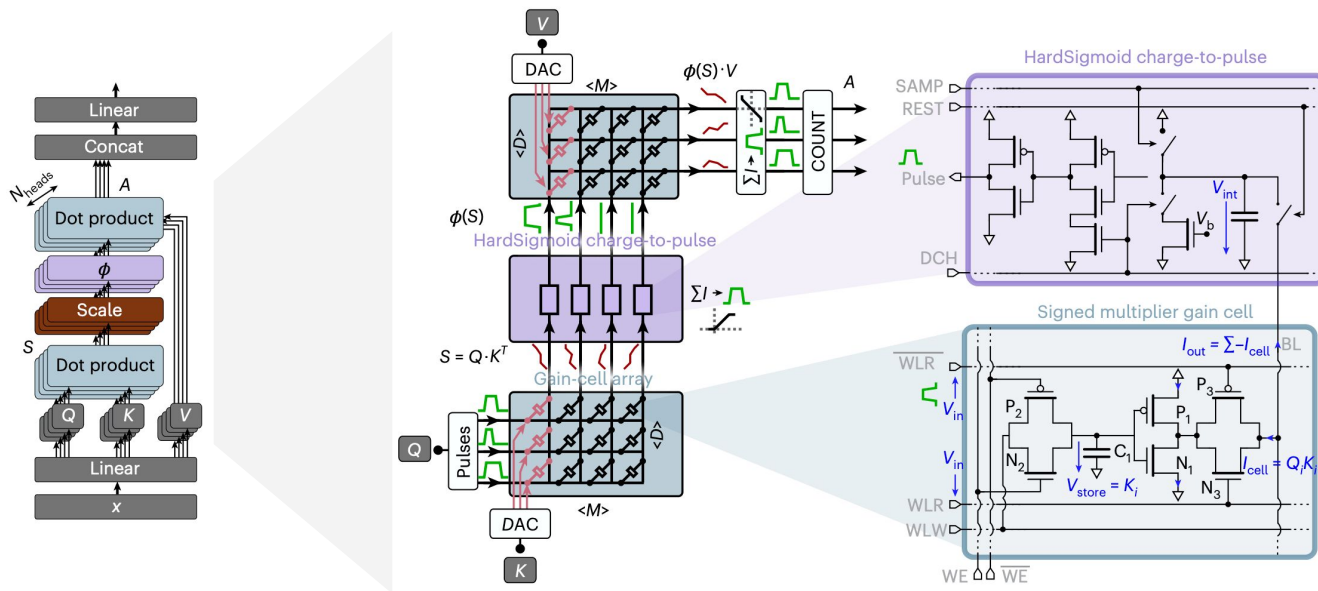
Impossible?

Putting together a perfectly filmed, coherent and insightful class lecture?

Ongoing challenges

Areas of improvement. Top of mind:

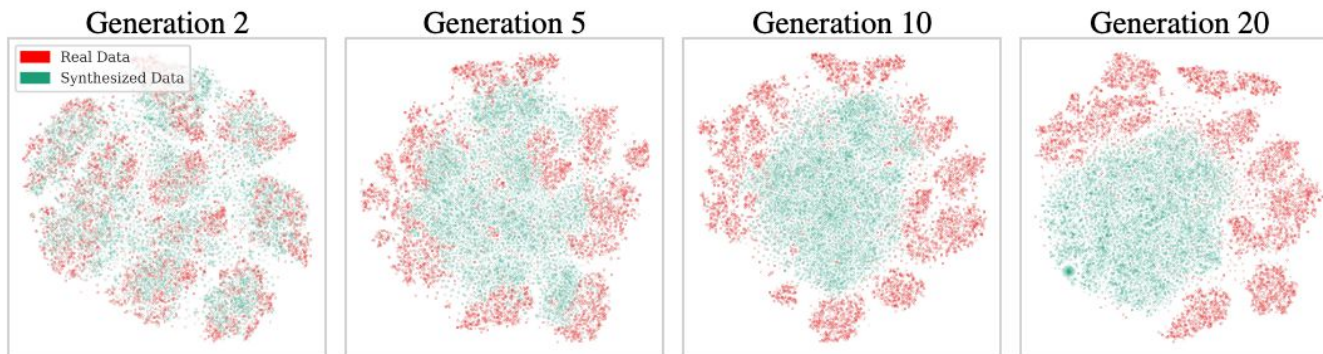
- Costs



Ongoing challenges

Areas of improvement. Top of mind:

- Costs
- Data quality



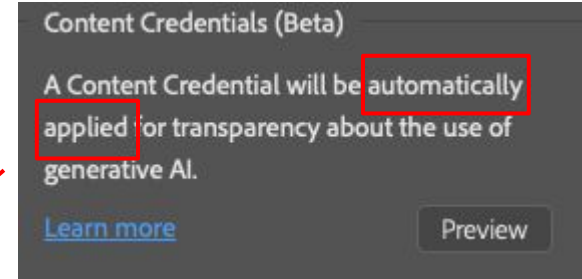
"Model collapse"

Ongoing challenges

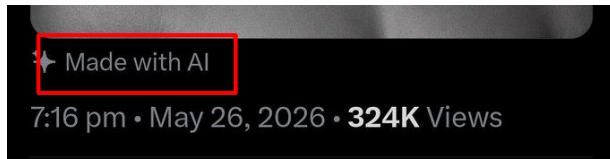
Areas of improvement. Top of mind:

- Costs
- Data quality
- Trust

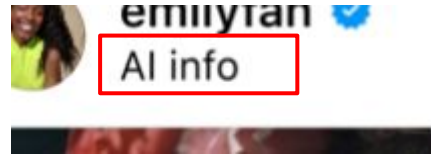
Trivial flaw: metadata stripping/screenshotting!



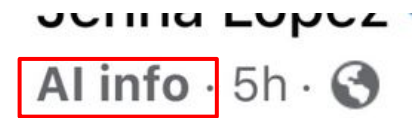
Photoshop



Twitter



Instagram



Facebook

Ongoing challenges

Areas of improvement. Top of mind:

- Costs
- Data quality
- Trust

Identity



Impulse noise



Rotation



Saturation



Watermarking

Ongoing challenges

Areas of improvement. Top of mind:

- Costs
- Data quality
- Trust
- Safety

PITTSBURGH'S
Public Source

TECHNOLOGY

**Deepfakes laws not moving at
the speed of AI**

How to stay up-to-date with Vision advances

Papers

- arXiv > Computer Science > Computer Vision and Pattern Recognition
- General ML (NeurIPS, ICML, ICLR) and Vision (CVPR, ECCV) venues

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Code

- Authors' GitHub repositories linked in their papers
- paperswithcode.com: browse state-of-the-art datasets/methods for each task

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Code

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- paperswithcode.com: browse state-of-the-art datasets/methods for each task

Miscellaneous

- Stanford course websites (e.g. [CS231n](#))
- Twitter (academics + industry leaders)
- YouTube theoretical ([Two Minute Papers](#), [Yannic Kilcher](#)) and practical ([Google Developers](#), [HuggingFace](#))
- Company/academia technical papers + blogs ([Amazon Science](#), [Apple ML](#), [Google AI](#), [Google DeepMind](#), [Microsoft Research](#), [OpenAI](#))

Maybe useful for after the class

VIP Cheatsheet

CME 296 – DIFFUSION & LARGE VISION MODELS

<https://cme296.stanford.edu>

VIP Cheatsheet: Diffusion & Large Vision Models

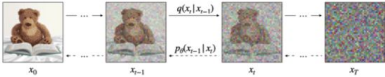
Afshine AMIDI and Shervine AMIDI
April 2, 2026

1 Generation paradigms

1.1 Discrete-time diffusion

Convention: Clean data is $t = 0$ and noisy data is $t = T$.

DDPM – A Denoising Diffusion Probabilistic Model (DDPM) learns to produce data via p_θ by reversing a known Markov diffusion process q that gradually adds noise to clean images.



Starting from a clean image $x_0 \sim p_0$, we obtain noisier images x_{t+1} using the current x_t with:

$$x_{t+1} = \sqrt{1 - \beta_t} x_t + \sqrt{\beta_t} \epsilon \quad \text{with } \beta_t \text{ noise schedule}$$

which can be rewritten as:

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon \quad \text{with } \epsilon \sim \mathcal{N}(0, 1), \alpha_t = 1 - \beta_t, \bar{\alpha}_t = \prod_{s=1}^t \alpha_s$$

KL divergence – Kullback-Leibler divergence (KL divergence) is a measure that compares two probability distributions p and q :

$$\text{KL}(p||q) = \int \log\left(\frac{p(z)}{q(z)}\right) p(z) dz = \mathbb{E}_p \left[\log\left(\frac{p}{q}\right) \right]$$

Variational formulation – Using the Evidence Lower Bound, we derive a tractable loss:

$$\mathbb{E}_{x_0}[\log(p_\theta(x_0))] \geq \text{ELBO} = - \sum_{t>1} \text{KL}(q(x_{t-1}|x_t, x_0) || p_\theta(x_{t-1}|x_t)) + \text{extra terms}$$

Derivation: Use Jensen's inequality and simplify loss to the KL divergence term.

Training – The DDPM loss estimates the amount of noise $\epsilon_\theta(x_t, t)$ to remove from the noisy image $x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon$ at time t by minimizing the L_2 loss with respect to the actual noise removed ϵ :

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{x_t, x_0, \epsilon} \left[\|\epsilon_\theta(x_t, t) - \epsilon\|^2 \right]$$

Sampling – We sample a less noisy image x_{t-1} using the noised image x_t and the predicted noise $\epsilon_\theta(x_t, t)$ to remove between x_t and x_0 :

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z$$

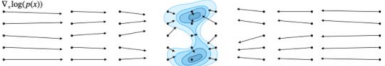
DDIM – A Denoising Diffusion Implicit Model (DDIM) is a deterministic sampling method to generate new samples faster by dropping the Markov assumption.

Derivation: Use convenient marginal distribution that has the same DDPM loss.

1.2 Score-based diffusion

Convention: Clean data is $t = 0$ and noisy data is $t = T$.

Score – The score function of a distribution is equal to $\nabla_x \log(p(x))$ and indicates the direction where the log-likelihood is the highest.



Remark: Compared to $\nabla_x p(x)$, the score is easier to compute, appears naturally in mathematical derivations and is more numerically stable, which is why it is used in practice.

DSM – Denoising Score Matching (DSM) is a training objective that learns the score function by predicting the gradient of the log data density from noisy inputs $x_\sigma = x + \sigma \epsilon$:



Training – Noise Conditional Score Networks (NCSN) trained with Score Matching with Langevin Dynamics (SMLD) estimates the score $\nabla_x \log(p_{\sigma_\tau}(x))$ at multiple noise levels σ_τ :

$$\mathcal{L}_{\text{NCSN}} = \sum_{\tau=1}^T \lambda_\tau \mathbb{E}_x \left[\|\epsilon_\theta(x, \sigma_\tau) - \nabla_x \log p_{\sigma_\tau}(x)\|_2^2 \right]$$

Sampling – Annealed Langevin Dynamics (ALD) is a sampling method that generates data while gradually reducing to progressively refine samples toward the data distribution:

$$x_t = x_{t-1} + \frac{\alpha_t}{2} s_\theta(x_{t-1}, \sigma_t) + \sqrt{\alpha_t} \epsilon_t$$

SDE formulation – The continuous formulation describes the forward diffusion process over time as a Stochastic Differential Equation (SDE) driven by a Wiener process, with time-dependent drift and diffusion terms that progressively perturb data with noise.

Thank you for this quarter!



Afshine

Shervine

A screenshot of the CME 296 course page on the Stanford University website. The page features the Stanford University logo, the course title "CME 296", and navigation links for "SYLLABUS", "CHEATSHEET", and "FAQ". The main heading is "CME 296 - Diffusion & Large Vision Models". Below this is a paragraph describing the course content. To the right, under the "Instructors" section, are circular portraits of Afshine Amidi and Shervine Amidi, both labeled as "Instructor". Under the "Logistics" section, it lists the time as "Fridays 3:30pm - 5:20pm", the location as "Thornton 110", and the grade as "Letter or Credit/No Credit". At the bottom, there are three buttons: "Syllabus", "Cheatsheet", and "Canvas".

 **CME 296** SYLLABUS CHEATSHEET FAQ

CME 296 - Diffusion & Large Vision Models

This course explores diffusion-based generative models for vision. You will study the foundations of diffusion, score matching and flow matching, modern architectures such as U-Nets and Diffusion Transformers, and methods for controllable image generation and evaluation. The course combines theory with practical insights into state-of-the-art generative models. Ideal for students with a background in linear algebra, probability, calculus and machine learning.

Instructors

 **Afshine Amidi**
Instructor

 **Shervine Amidi**
Instructor

Logistics

Time: Fridays 3:30pm - 5:20pm
Location: Thornton 110
Grade: Letter or Credit/No Credit

[Syllabus](#) [Cheatsheet](#) [Canvas](#)

CME 296: Diffusion & Large Vision Models

Website: <https://cme296.stanford.edu/>

See you this Fall?

 **CME 295**

[SYLLABUS](#) [CHEATSHEET](#) [FAQ](#)

CME 295 - Transformers & Large Language Models

This course explores the world of Transformers and Large Language Models (LLMs). You'll learn the evolution of NLP methods, the core components of the Transformer architecture, along with how they relate to LLMs as well as techniques to enhance model performance for real-world applications. Through a mix of theory and practical insights, this course will equip you with the knowledge to leverage LLMs effectively. Ideal for those with a background in calculus, linear algebra, and basic machine learning concepts.

[Syllabus](#) [Cheatsheet](#) [Canvas](#)

Instructors

**Afshine Amidi**
Instructor

**Shervine Amidi**
Instructor

Logistics

Time: Fridays 3:30pm - 5:20pm
Location: Thornton 110
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CME 295: Transformers & Large Language Models

Website: <https://cme295.stanford.edu/>



Wishing you all the best!

Feel free to reach out if you want to chat!